

INDEX OF CONSTRUCTION AND MATERIAL SPECIFICATIONS

SPEC NO.	CONSTRUCTION SPECIFICATION	SPEC NO.	CONSTRUCTION SPECIFICATION
4	Channel Clearing and Shaping	94	Contractor Quality Control
5	Pollution Control	95	Geotextile
6	Seeding, Sprigging and Mulching		
8	Mobilization and Demobilization		
9	Traffic Control		
11	Removal of Water		
13	Piling	SPEC NO.	MATERIAL SPECIFICATION
21	Excavation	511	Steel Piles
23	Earthfill	521	Aggregates for Drainfill and Filters
24	Drainfill	523	Rock for Riprap
45	Plastic Pipe	547	Plastic Pipe
61	Rock Riprap	592	Geotextile

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Construction Specification 4—Channel Clearing and Shaping

1. Scope

The work consists of clearing designated areas by the removal and disposal of trees, logs, stumps, shrubs, brush, and rubbish, and the shaping of the channel.

2. Protection of existing vegetation

Trees and other vegetation designated to remain undisturbed shall be protected from damage throughout the duration of the construction period. Any damages resulting from the contractor's operations or neglect shall be repaired by the contractor.

Earthfill, stockpiling of materials, vehicle or equipment parking, and excessive foot or vehicle traffic shall not be allowed within the drip line of vegetation designated to remain in place. Vegetation damaged by any of these or similar actions shall be replaced with viable vegetation of the same species, similar condition, and like size unless otherwise approved by the contracting officer.

Any cuts, skins, scrapes, or bruises to the bark of the vegetation shall be carefully trimmed and local nursery accepted procedures used to seal damaged bark.

Any limbs or branches 0.5 inch or larger in diameter that are broken, severed, or otherwise seriously damaged during construction shall be cut off at the base of the damaged limb or branch flush with the adjacent limb or tree trunk.

All roots 1 inch or larger in diameter that are cut, broken, or otherwise severed during channel shaping shall have the end smoothly cut perpendicular to the root. Roots exposed during channel shaping operations shall be covered with moist soil as soon as possible to prevent roots from drying out.

3. Marking

The limits of the area(s) to be cleared and shaped are marked by stakes, flags, paint, tree markings, or other suitable methods or as specified in section 8 of this specification, or they will be shown on the drawings. Trees to remain standing, undisturbed, and uninjured are designated by special markings.

4. Clearing

Trees and other vegetation marked for clearing shall be cut off as near the ground surface as conventional tools and equipment normally permit. All trees not marked for preservation and all snags, logs, brush, shrubs, stumps, and rubbish shall be cleared from within the area limits identified.

5. Disposal

All woody material, vegetation, and rubbish resulting from clearing from designated areas shall be disposed of at the locations and in a manner shown on the drawings, or as specified in section 8 of this specification.

6. Shaping

The channel bottom and side slopes shall be shaped as shown on the drawings. The resulting shaped channel surface shall be reasonably smooth. Material excavated during the channel shaping operation shall be removed from the channel and disposed of as specified in section 8 of this specification.

7. Measurement and payment

Method 1—For items of work for which specific unit prices are established by the contract, the designated cleared and shaped area is measured and the area determined to the nearest 0.1 acre. Payment for clearing and shaping is made at the contract unit price for the item and shall constitute full compensation for all labor, equipment, tools, applicable permits and associated fees for burning and disposal of refuse, and all other items necessary and incidental to the satisfactory completion of the work.

Method 2—For items of work for which specific unit prices are established by the contract, the length of the cleared and shaped channel designated is measured to the nearest 100 feet. Payment for clearing and shaping is made at the contract unit price for the item and shall constitute full compensation for all labor, equipment, tools, applicable permits and associated fees for burning and disposal of refuse, and all other items necessary and incidental to the satisfactory performance of the work.

Method 3—For items of work for which specific unit prices are established by the contract, the cleared and shaped area(s) is measured and the area determined to the nearest 0.1 acre. The designated cleared and shaped area(s) is determined from the measured width at representative sections and the distance between the sections. Payment for clearing and shaping is made at the contract unit price for the item and shall constitute full compensation for all labor, equipment, tools, applicable permits and associated fees for burning and disposal of refuse, and all other items necessary and incidental to the satisfactory completion of the work.

Method 4—For items of work for which specific lump sum prices are established by the contract, the extent of clearing and shaping is not measured or determined for payment. Payment for clearing and shaping is made at the contract lump sum price for the item and shall constitute full compensation for all labor, equipment, tools, applicable permits and associated fees for burning and disposal of refuse, and all other items necessary and incidental to the satisfactory completion of the work.

All Methods—The following provisions apply to all methods of measurement and payment. Compensation for any item of work described in the contract, but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and items to which they have been made subsidiary are identified in section 8 of this specification.

8. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details therefore are:

a. Bid Item 2 - Channel Clearing and Shaping

1. This item shall consist of removal and disposal of trees, logs, stumps, brush, and rubbish along the streambank and the shaping of the channel bank as shown and described in the drawings.
2. All removed woody material and rubbish shall be disposed off-site at an approved dump site. No on-site burning or burying will be authorized.
3. Excess or unsuitable material encountered during channel clearing and shaping such as soils unsuitable for use as fill material, organic material such as roots, or other foreign materials shall not be spoiled or graded within the floodplain. Excess or unsuitable materials shall be disposed off site at an environmentally responsible site approved by the COR.
4. All trees and rootwads shall be removed from within the confines of the proposed steel sheet pile wall and the tie-backs. Any trees designated as not to be disturbed, shall be marked with ribbons.
5. The beginning of the clearing and shaping will be from approximately station 0+28 to approximately station 3+04.
6. Fill material placed in front of the sheet pile wall shall be obtained from the excavation of drainfill and riprap areas or sourced from an off site location. Fill material shall be capable of supporting vegetation. Fill shall be placed to the lines and elevations indicated by the drawings.
7. Measurement and payment shall be by Method 4.
8. Items subsidiary to this bid item are:
 - a. Seeding and Mulching - Construction Specification - 6
 - b. Removal of Water - Construction Specification 11
 - c. Excavation - Construction Specification - 21
 - d. Earthfill - Construction Specification - 23

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Construction Specification 5—Pollution Control

1. Scope

The work consists of installing measures or performing work to control erosion and minimize the production of sediment and other pollutants to water and air from construction activities.

The following BioPreferred® product categories are applicable to this specification: — mulch and compost materials, erosion control materials, fertilizers, dust suppressants, agricultural spray adjuvants

2. Material

Silt fence shall conform to the requirement of Materials Specification 592, Geotextile. All other material furnished shall meet the requirements of the material specifications listed in section 8 of this specification.

3. Erosion and sediment control measures and works

The measures and works shall include, but are not limited to, the following:

Staging of earthwork activities—The excavation and moving of soil materials shall be scheduled to minimize the size of areas disturbed and unprotected from erosion for the shortest reasonable time.

Seeding—Seeding to protect disturbed areas shall occur as soon as reasonably possible following completion of that earthwork activity.

Mulching—Mulching to provide temporary protection of the soil surface from erosion.

Diversions—Diversions to divert water from work areas and to collect water from work areas for treatment and safe disposition. They are temporary and shall be removed and the area restored to its near original condition when the diversions are no longer required or when permanent measures are installed.

Stream crossings—Culverts or bridges where equipment must cross streams. They are temporary and shall be removed and the area restored to its near original condition when the crossings are no longer required or when permanent measures are installed.

Sediment basins—Sediment basins collect, settle, and eliminate sediment from eroding areas from impacting properties and streams below the construction site(s). These basins are temporary and shall be removed and the area restored to its original condition when they are no longer required or when permanent measures are installed.

Sediment filters—Straw bale filters or geotextile silt fences trap sediment from areas of limited runoff. Sediment filters shall be properly anchored to prevent erosion under or around them. Silt fences shall be installed and maintained in accordance with ASTM D6462. These filters are temporary and shall be removed and the area restored to its original condition when they are no longer required or when permanent measures are installed.

Waterways—Waterways for the safe disposal of runoff from fields, diversions, and other structures or measures. These works are temporary and shall be removed and the area restored to its original condition when they are no longer required or when permanent measures are installed.

Other—Additional protection measures as specified in section 8 of this specification or required by Federal, State, or local government.

4. Chemical pollution

The contractor shall provide watertight tanks or barrels or construct a sump sealed with plastic sheets to collect and temporarily contain chemical pollutants, such as drained lubricating or transmission fluids, grease, soaps, concrete mixer washwater, or asphalt, produced as a by-product of the construction activities. Pollutants shall be disposed of in accordance with appropriate state and Federal regulations. At the completion of the construction work, tanks, barrels, and sumps shall be removed and the area restored to its original condition as specified in section 8 of this specification. Sump removal shall be conducted without causing pollution.

Sanitary facilities, such as chemical toilets, or septic tanks shall not be located next to live streams, wells, or springs. They shall be located at a distance sufficient to prevent contamination of any water source. At the completion of construction activities, facilities shall be disposed of without causing pollution as specified in section 8 of this specification.

5. Air pollution

The burning of brush or slash and the disposal of other materials shall adhere to state and local regulations.

Fire prevention measures shall be taken to prevent the start or spreading of wildfires that may result from project activities. Firebreaks or guards shall be constructed and maintained at locations shown on the drawings.

All public access or haul roads used by the contractor during construction of the project shall be sprinkled or otherwise treated to fully suppress dust. All dust control methods shall ensure safe construction operations at all times. If chemical dust suppressants are applied, the material shall be a commercially available product specifically designed for dust suppression and the application shall follow manufacturer's requirements and recommendations. A copy of the product data sheet and manufacturer's recommended application procedures shall be provided to the engineer 5 working days before the first application.

6. Maintenance, removal, and restoration

All pollution control measures and temporary works shall be adequately maintained in a functional condition for the duration of the construction period. All temporary measures shall be removed and the site restored to near original condition.

7. Measurement and payment

Method 1—For items of work for which specific unit prices are established in the contract, each item is measured to the nearest unit applicable. Payment for each item is made at the contract unit price for that item. For water or chemical suppressant items used for dust control for which items of work are established in section 8 of this specification, measurement for payment will not include water or chemical suppressants that are used inappropriately or excessive to need. Such payment will constitute full compensation for the completion of the work.

Method 2—For items of work for which lump sum prices are established in the contract, payment is made as the work proceeds and supported by invoices presented by the contractor that reflect actual costs. If the total of all progress payments is less than the lump sum contract price for this item, the balance remaining for this item will be included in the final contract payment. Payment of the lump sum contract price will constitute full compensation for completion of the work.

Method 3—For items of work for which lump sum prices are established in the contract, payment will be prorated and provided in equal amounts on each monthly progress payment estimate. The number of months used for prorating shall be the number estimated to complete the work as outlined in the contractor's approved construction schedule. The final month's prorate amount will be provided with the final contract payment. Payment as described will constitute full compensation for completion of the work.

All Methods—The following provisions apply to all methods of measurement and payment. Compensation for any item of work described in the contract, but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items, and the items to which they are made subsidiary, are identified in section 8 of this specification.

8. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details therefore are:

Subsidiary Item - Pollution Control

1. The Contractor is required to provide the Contracting Officer's Representative with a written Pollution Control Plan of Operations for this project. The plan shall be submitted prior to the pre-construction conference, and reviewed by all parties during the pre-construction conference.
2. The Contractor shall perform the work of this project in a manner consistent and in conformance with Federal, State, or Local Government regulations, to protect the site and downstream areas from degradation.
3. Method of Payment.
Section 7 Measurement and Payment does not apply. No separate payment will be made for Pollution Control. Compensation for this item will be included in the payment for Bid Item 1 - Mobilization / Demobilization.

Construction Specification 6—Seeding, Sprigging, and Mulching

1. Scope

The work consists of preparing the area for treatment; furnishing and placing seed, sprigs, mulch, fertilizer, inoculant, lime, and other soil amendments; and anchoring mulch in designated areas as specified.

The following BioPreferred® product categories are applicable to this specification:

- Mulch and compost materials
- Erosion control materials
- Fertilizers
- Agricultural spray adjuvants

2. Material

Seed—All seed must conform to the current rules and regulations of the State where it is being used and must be from the latest crop available. It must meet or exceed the standard for purity and germination listed in section 7.

Seed must be labeled in accordance with State laws and U.S. Department of Agriculture rules and regulations under the Federal Seed Act in effect on the date of invitations for bids. Bag tag figures are evidence of purity and germination. No seed may be accepted with a test date of more than 9 months before the date of delivery to the site.

Seed that has become wet, moldy, or otherwise damaged in transit or storage will not be accepted. The percent of noxious weed seed allowable must be as defined in the current State laws relating to agricultural seeds. Each type of seed must be delivered in separate sealed containers and fully tagged unless an exception is granted in writing by the contracting officer.

Fertilizer—Unless otherwise specified, the fertilizer must be a commercial-grade fertilizer. It must meet the standard for grade and quality specified by State law. Where fertilizer is furnished from bulk storage, the contractor must furnish a supplier's certification of analysis and weight. When required by the contract, a representative sample of the fertilizer must be furnished to the contracting officer for chemical analysis.

Inoculants—The inoculant for treating legume seeds must be a pure culture of nitrogen-fixing bacteria prepared specifically for the species and must not be used later than the date indicated on the container or as otherwise specified. A mixing medium, as recommended by the manufacturer, must be used to bond the inoculant to the seed. Two times the amount of the inoculant recommended by the manufacturer must be used, except that four times the amount must be used when seed is applied using a hydraulic seeder. Seed must be sown within 24 hours of treatment and must not remain in the hydraulic seeder longer than 4 hours.

Lime and other soil amendments—Lime must consist of standard ground agriculture limestone, or approved equivalent. Standard ground agriculture limestone is defined as ground limestone meeting current requirements of the State department of agriculture. Other soil amendments must meet quality criteria and application requirements specified in section 7.

Mulch tackifiers—Asphalt emulsion tackifiers must conform to the requirements of ASTM D977, Specification for Emulsified Asphalt. The emulsified asphalt may be rapid setting, medium setting, or slow setting. Non-asphaltic tackifiers required because of environmental considerations must be as specified in section 7.

Straw mulch material—Straw mulch must consist of wheat, barley, oat or rye straw, hay, grass cut from native grasses, or other plants as specified in section 7. The mulch material must be air-dry, reasonably light in color, and must not be musty, moldy, caked, or otherwise of low quality. The use of mulch that contains noxious weeds is not permitted. The contractor must provide a method satisfactory to the contracting officer for determining weight of mulch furnished.

Other mulch materials—Mulching materials, such as wood cellulose fiber mulch, mulch tackifiers, synthetic fiber mulch, netting, and mesh may be required for specialized locations and conditions. These materials, when specified, must be accompanied by the manufacturer's recommendations for methods of application.

3. Seeding mixtures, sod, sprigs, and dates of planting

The application rate per acre for seed mixtures, sprigs, or sod and date of seeding or planting must be as shown on the plans or as specified in section 7.

4. Seedbed preparation and treatment

Areas to be treated must be dressed to a smooth, firm surface. On sites where equipment can operate on slopes safely, the seedbed must be adequately loosened (4 to 6 inches deep) and smoothed. Depending on soil and moisture conditions, disking or culti-packing, or both, may be necessary to properly prepare a seedbed. Where equipment cannot operate safely, the seedbed must be prepared by hand methods by scarifying to provide a roughened soil surface so that broadcast seed will remain in place.

If seeding is to be accomplished immediately following construction operations, seedbed preparation may not be required except on a compacted, polished, or freshly cut soil surface.

Rocks larger than 6 inches in diameter, trash, weeds, and other debris that will interfere with seeding or maintenance operations must be removed or disposed of as specified in section 7.

Seedbed preparation must be discontinued when soil moisture conditions are not suitable for the preparation of a satisfactory seedbed as determined by the responsible engineer.

5. Seeding, sprigging, fertilizing, mulching, and stabilizing

All seeding or sprigging operations must be performed in such a manner that the seed or sprigs are applied in the specified quantities uniformly in the designated areas. The method and rate of seed application must be as specified in section 7. Unless otherwise specified, seeding or sprigging must be accomplished within 2 days after final grading is completed and approved.

Fertilizer, lime, and other soil amendments must be applied as specified in section 7.

When specified, the fertilizer and soil amendments must be thoroughly incorporated into

the soil immediately following surface application.

The rate, amount, and kind of mulching or mesh must be as specified in section 7. Mulches must be applied uniformly to the designated areas. They must be applied to areas seeded not later than 2 working days after seeding has been performed. Straw mulch material must be stabilized within 24 hours of application using a mulch crimper or equivalent anchoring tool or by a suitable tackifier. When the mulch crimper or equivalent anchoring tool is used, it must have straight blades and be the type manufactured expressly for and capable of firmly punching the mulch into the soil. Where the equipment can be safely operated, it must be operated on the contour. Hand methods must be used where equipment cannot safely operate to perform the work required.

The tackifier must be applied uniformly over the mulch material at the specified rate, or it must be injected into the mulch material as it is being applied. Mesh or netting stabilizing materials must be applied smoothly but loosely on the designated areas. The edges of these materials must be buried or securely anchored using spikes or staples as specified in section 7.

The contractor must maintain the mesh or netting areas until all work under the contract has been completed and accepted. Maintenance must consist of the repair of areas damaged by water erosion, wind, fire, or other causes. Such areas must be repaired to reestablish the intended condition and to the design lines and grades required by the contract. The areas must be refertilized, reseeded, and remulched before the new application of the mesh or netting.

6. Measurement and payment

Method 1—For items of work for which specific unit prices are established in the contract, each area treated is measured as specified in section 7 and the area is calculated to the nearest 0.1 acre. Payment for treatment is made at the contract unit price for the designated treatment, which will constitute full compensation for completion of the work.

When specified as an item of work, mesh or netting is measured to the nearest square yard of surface area covered and accepted. Payment is made at the contract unit price and will constitute full compensation for completion of the work.

Method 2—For items of work for which specific lump-sum prices are established in the contract, the quantity of work will not be measured for payment. Payment for this item is made at the contract lump-sum price for the item and constitutes full compensation for the completion of the work.

Method 3—For items of work for which lump-sum prices are established in the contract, payment is made as the work proceeds. Progress payments are determined as specified in section 7. Payment of the lump sum contract price constitutes full compensation for completion of the work.

All Methods—The following provisions apply to all methods of measurement and payment. Compensation for any item of work described in the contract but not listed in

the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in section 7.

7. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details therefore are:

Subsidiary Item- Seeding, Sprigging and Mulching

This item shall include seeding and mulching of all disturbed areas within the construction limits as shown and described in the drawings.

1. Fertilizer

All areas shall have fertilizer uniformly applied at the following rates:

- Nitrogen (N) - 150 pounds per acre
- Phosphorus P₂O₅ - 150 pounds per acre
- Potassium K₂O - 150 pounds per acre

This specification can be met by applying fertilizer having an analysis of 15-15-15 at the rate of 1000 pounds per acre or 23 pounds per thousand square feet.

2. Seed

All seed shall be labeled, tagged or marked in accordance with Section 907.03 of the Revised Code of Ohio. Seed tags showing germination and purity shall be attached to all bags. No pre-mixed seed shall be used unless the following conditions are met:

a. The weight, purity, and germination for each kind of seed is specified on the tag for each lot of seed. Seeding rates shall be adjusted for Pure Live Seed (PLS) at the time of seeding. Pure Seed (PLS) = percent germination. Seed shall have the equivalent of a minimum of 80% pure live seed.

When the percent purity multiplied by the percent germination gives a percentage less than 80, the rate of seeding will be increased to provide a minimum PLS of 80.

b. Seeding mixtures and rates per acre:

Disturbed construction areas -

Kentucky 31 Tall Fescue -60 lbs per acre or 1.5 lbs per 1000 square ft.

Annual Ryegrass -10 lbs per acre or 0.25 lbs per 1000 square ft.

Streambank area (riprap toe to pile wall):

Kentucky 31 Tall Fescue -60 lbs per acre or 1.5 lbs per 1000 square ft.

Annual Ryegrass -10 lbs per acre or 0.25 lbs per 1000 square ft.

Wild Rye (*Elymus riparius*)-20 lbs per acre or 0.5 lbs per 1000 square ft.

Little Blue Stem -10lbs per acre or 0.25 lbs per 1000 square ft.

3. Seedbed Preparation

Seedbed preparation shall be completed within 48 hours after application of fertilizer. All areas accessible to equipment shall be disked to incorporate fertilizer and to prepare a seedbed a minimum of three inches deep. The final seedbed preparation operation shall be performed across the slope and immediately prior to seeding. Suitable equipment shall be used to smooth the seedbed prior to seeding. On areas inaccessible to a disk, a one-inch minimum seedbed will be prepared using suitable tillage tools or hand tools. All rocks greater than 3 inches in size shall be removed.

4. Sowing the Seed

Seeding shall be completed as soon as practical after final grading. Broadcast the seed and cover approximately 1/4-inch deep with a light harrow, culti-packer, or other suitable equipment, operated across the slope. On areas too steep or otherwise inaccessible to equipment, the seed shall be covered with a hand rake, or other suitable equipment.

5. Mulch and Method of Anchoring

Mulch shall consist of (wheat, rye, or barley) straw and shall be applied uniformly at 2 air-dried tons per acre or 100 lbs. per 1000 square feet. Mulch shall be applied within 48 hours after application of seeding. All bailing wire and twine shall be removed from the seeded and mulched areas. No anchoring is required.

6. Erosion Control Blanket

Erosion control blanket shall be installed along the stream bank as shown in the drawings. Erosion control blankets shall be double-netted using a mulch medium comprised of straw, coconut fiber, or equivalent material. Erosion control blankets shall have a minimum dry weight per surface area of 0.5 lbs/sq. yd as per ASTM D6475. Erosion control blankets shall have a 12-month function longevity as described in FWHA FP-14. Minimum 6 inch long staples shall be installed per the stapling plan shown in the drawings.

7. Method of Payment

Section 6 Measurement and Payment does not apply. No separate payment will be made for Seeding, Sprigging, and Mulching. Compensation for this item will be included in the payment for Bid Item 2 – Channel Clearing and Shaping.

Construction Specification 8—Mobilization and Demobilization

1. Scope

The work consists of the mobilization and demobilization of the contractor's forces and equipment necessary for performing the work required under the contract. It does not include mobilization and demobilization for specific items of work for which payment is provided elsewhere in the contract. Mobilization will not be considered as work in fulfilling the contract requirements for commencement of work.

2. Equipment and material

Mobilization shall include all activities and associated costs for transportation of contractor's personnel, equipment, and operating supplies to the site; establishment of offices, buildings, and other necessary general facilities for the contractor's operations at the site; premiums paid for performance and payment bonds including coinsurance and reinsurance agreements as applicable; and other items specified in section 4 of this specification.

Demobilization shall include all activities and costs for transportation of personnel, equipment, and supplies not required or included in the contract from the site; including the disassembly, removal, and site cleanup of offices, buildings, and other facilities assembled on the site specifically for this contract.

This work includes mobilization and demobilization required by the contract at the time of award. If additional mobilization and demobilization activities and costs are required during the performance of the contract as a result of changed, deleted, or added items of work for which the contractor is entitled to an adjustment in contract price, compensation for such costs will be included in the price adjustment for the item or items of work changed or added.

3. Payment

Payment will be made as the work proceeds, after presentation of paid invoices or documentation of direct costs by the contractor showing specific mobilization and demobilization costs and supporting evidence of the charges of suppliers, subcontractors, and others. When the total of such payments is less than the lump sum contract price, the balance remaining will be included in the final contract payment. Payment of the lump sum contract price for mobilization and demobilization will constitute full compensation for completion of the work.

Payment will not be made under this item for the purchase costs of materials having a residual value, the purchase costs of materials to be incorporated in the project, or the purchase costs of operating supplies.

4. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details therefore are:

Bid Item 1 - Mobilization and Demobilization

1. This item shall consist of mobilizing and demobilizing all forces, equipment, and materials necessary for performing the work required under this contract.
2. Any ingress/egress or storage/lay-down areas within the Work Limits must be returned to the condition existing prior to construction. Any storage/lay-down areas outside the Work Limits shall be coordinated between the Contractor and Landowner.
3. A written and detailed construction schedule, prepared by the Contractor, will be required prior to beginning work.
4. A portable toilet shall be provided by the Contractor. The portable toilet shall be placed within the Work Limits and shall be operated and maintained by the Contractor for the duration of the project.
5. Approximately 2,000 ft north of the project site, Blessing Rd crosses Raccoon Creek via bridge at 38°46'03.1"N 82°18'41.0"W. No construction traffic shall utilize the Blessing Rd bridge for ingress or egress from the construction site. All construction traffic shall use alternative routes via Smokey Row Rd.
6. The contractor shall notify Gallia Rural Water via phone at 740-446-9221 at minimum 2 working days prior to the start of construction and at minimum 2 working days prior to the start of sheet pile vibration.
7. The contractor shall contact Buckeye Rural Electric Coop at minimum 30 days prior to the start of construction to coordinate the temporary relocation of the overhead power line. Coordination with Buckeye Rural Electric Coop shall be routed through Matthew Nibert (740-379-9638). Expenses billed from Buckeye Rural Electric Coop pertaining to the temporary relocation of the power line are the responsibility of the contractor and shall be included in this bid item.
8. All concrete barriers and reflectors installed along Blessing Rd shall be removed prior to the start of construction and stockpiled within the work limits for later collection by the Green Township trustees.

9. Matthew Smith, Assistant Regional Scenic River Manager, shall be invited to the preconstruction meeting between the contractor and NRCS. Matthew Smith shall be notified of the project start date one week prior to prior to the commencement of work. Periodic inspection of the project may take place. Matthew Smith shall be contacted one week prior to the completion of the project. Matthew Smith can be contacted at matthew.smith@dnr.ohio.gov and/or 440-225-5582.
10. Items subsidiary to this bid item are:
 - a. Pollution Control -Construction Specification 5
 - b. Traffic Control -Construction Specification 9
 - c. Contractor Quality Control -Construction Specification 94

Construction Specification 9—Traffic Control

1. Scope

The work shall consist of establishing traffic control and maintaining safe, convenient use of public roads and rights-of-way.

2. Traffic and access

The contractor's operations shall cause no unnecessary inconvenience to the public. The public rights-of-way shall be maintained at all times unless interruption is authorized by proper local authority. Contractor's authorized closing or detour plans shall be provided to the engineer for approval.

Safe and adequate access shall be provided and maintained to all public protection devices and to all critical utility control locations. Facility access shall be continuous and unobstructed unless otherwise approved.

3. Storage of equipment and material in public streets

Construction materials and equipment shall not be stored or parked on public streets, roads, or highways. During any material or equipment loading or unloading activities that may temporarily interfere with traffic, an acceptable detour shall be provided for the duration of the activity. Any associated expense for this activity is the responsibility of the contractor.

Excavated material, including suitable material that is intended for adjacent trench backfill or other earth backfill as specified in section 5 of this specification, shall not be stored on public streets, roads, or highways that remain in service for the public. Any waiver of this requirement must be obtained from the proper local authority and approved by the engineer. All excess and unsuitable material shall be removed from the site as soon as possible. Any spillage shall be removed from roadways before they are used by the public.

4. Street closures, detours, and barricades

The contractor shall comply with the requirements of all applicable responsible units of government for closure of any street, road, or highway. The contractor shall provide the required barriers, guards, lights, signs, temporary bridges, and flaggers together with informing the public of any detours and construction hazards by the most suitable means available, such as local newspapers or radio stations. The contractor is also responsible for compliance with additional public safety requirements that may arise during construction. The contractor shall furnish, install, and, upon completion of the work, promptly remove all signs, warning devices, and other materials used in the performance of this work.

Unless otherwise specified, the contractor shall notify, in writing, the fire chief, police chief, county sheriff, state patrol, schools that operate school buses, or any other government official as may be appropriate no less than 7 days before closing, partly closing, or reopening any street, road, or highway.

Unless otherwise specified, the contractor shall furnish to the engineer a written plan

showing the proposed method of signing, barricading for traffic control, and safety for street detours and closures.

All temporary detours will be maintained to ensure use of public rights-of-way is provided in a safe manner. This may include dust control, grading, and graveling as required in section 7 of this specification.

5. General and specific references

All signs, signals, barricades, use of flaggers, and other traffic control and public safety devices shall conform to the general requirements set forth in the Manual of Uniform Traffic Control Devices (MUTCD) and the latest edition of *Standard Highway Signs and Standard Alphabets for Highway Signs* and/or *OSHA Construction Industry Standards (29 CFR Part 1926), Subpart G, Signs, Signals, and Barricades* unless otherwise specified in section 7 of this specification.

6. Measurement and payment

For items of work for which specific lump sum prices are established in the contract, payment for the work is made at the contract lump sum price. Progress payments will be made based upon the percentage of estimated total time that traffic control will be required unless otherwise specified in section 7 of this specification. Payment will constitute full compensation for all flaggers, labor, materials, equipment, and all other items necessary and incidental to completion of the work.

Compensation for any item of work described in the contract, but not listed in the bid schedule will be included in the payment for the item of work to which it is made subsidiary. Such items and items to which they are made subsidiary are identified in section 7 of this specification.

7. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details therefore are:

Subsidiary Item - Traffic Control

1. Blessing Road may be closed to through traffic during the period of construction. The proposed road closure restrictions and maintenance of traffic requirements are shown in the drawings. The Contractor shall contact the Green Township Trustees to arrange road closure and additional signage required. Notification to the Trustees shall be made at least 3 working days prior to the start of any road closure, to allow the Trustees sufficient time to notify emergency & other services of the closure.
 - Contact the Green Township Trustees at (740) 379-2580.
2. Road closures shall be in accordance with ODOT Construction and Material Specifications Item 614.05 and other applicable portions of the specifications, as well as the Ohio Manual of Uniform Traffic Control Devices. A written Traffic Control & Road Closure Plan shall be submitted to the NRCS engineer at the pre-construction conference.
3. Road closures barriers and signage shall be posted at minimum at the locations shown in the drawings. Access to Wray Rd and Smokey Row Rd, with adequate space for traffic turnaround, shall be maintained during the closure of Blessing Rd.
4. Local traffic access shall be maintained for 2050 Blessing Rd and 2100 Blessing Rd from the Blessing Rd bridge and Wray Rd. Local traffic access shall be maintained for 2430 Blessing Rd from Smokey Row Rd.
5. Access to the barn at 2430 Blessing Rd, 484' south of BM2 west of Blessing Rd, shall be maintained for the landowner throughout the construction period.
6. Method of Payment
Section 6 Measurement and Payment does not apply. No separate payment will be made for Traffic Control. Compensation for this item will be included in the payment for Bid Item 1 - Mobilization and Demobilization.

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Construction Specification 11—Removal of Water

1. Scope

The work consists of the removal of surface water and ground water as necessary to perform the construction required by the contract in accordance with the specifications. It shall include: (1) constructing, installing, building, and maintaining all necessary temporary water containment facilities, channels, and diversions; (2) furnishing, installing, and operating all necessary pumps, piping, and other facilities and equipment; and (3) removing all such temporary works and equipment after their intended function is no longer required.

2. Diverting surface water

The contractor shall install, maintain, and operate all cofferdams, channels, flumes, sumps, and all other temporary diversion and protective works needed to divert streamflow and other surface water through or around the construction site. Control of surface water shall be continuous during the period that damage to construction work could occur. Unless otherwise specified and/or approved, the diversion outlet shall be into the same drainageway that the water would have reached before being diverted.

The contractor shall furnish the contracting officer, in writing, a proposed plan for diverting surface water before beginning any construction activities for which a diversion is required, unless waived in section 8 of this specification. Acceptance of this plan or the waiving of the plan requirement will not relieve the contractor of the responsibilities related to this activity during the process of completing the work as specified.

3. Dewatering the construction site

Foundations, cutoff trenches, and all other parts of the construction site shall be dewatered and kept free of standing water and muddy conditions as necessary for the proper execution of the work. The contractor shall furnish, install, operate, and maintain all drains, sumps, pumps, casings, well points, and all other equipment required to properly dewater the site as specified. Dewatering systems that cause a loss of soil fines from the foundation areas will not be permitted.

The contractor shall furnish the contracting officer, in writing, a proposed plan for dewatering before commencing with any construction activity for which dewatering may be required, unless waived in section 8 of this specification. Acceptance of this plan or the waiving of the plan requirement will not relieve the contractor of the responsibilities for completing the specified work.

4. Dewatering borrow areas

The contractor shall maintain all borrow areas free of surface water or otherwise provide for timely and effective removal of surface and subsurface water that accumulates within the borrow area, unless waived in section 8 of this specification. Borrow material shall be processed as necessary to achieve proper and uniform moisture content at the time of placement.

If pumping to dewater borrow areas is included as a bid item of work in the bid schedule, each pump discharge pipe shall be equipped with a water meter. The meter shall be such that the measured quantity of water is accurate within 3 percent of the true

quantity. The contractor shall provide necessary support to perform accuracy tests of the water meter when requested by the contracting officer.

5. Erosion and pollution control

Removal of water from the construction site, including the borrow areas, shall be accomplished so that erosion and the transporting of sediment and other pollutants are minimized. Dewatering activities shall be accomplished in a manner that the water table water quality is not altered. Pollution control activities shall not conflict with the requirements of Construction Specification 5, Pollution Control, if it is a part of this contract.

6. Removal of temporary works

When temporary works are no longer needed, the contractor shall remove and return the area to a condition similar to that which existed before construction. Areas where temporary works were located shall be graded for slightly appearance with no obstruction to natural surface waterflows or the proper functioning and access to the works of improvement installed. The contractor shall exercise extreme care during the removal stages to minimize the loss of soil sediment and debris that was trapped during construction.

Pipes, casings, and any other material used to dewater the site shall be removed from temporary wells. The wells shall be filled to ground level with clean gravel or other suitable material approved by the contracting officer. The contractor shall exercise extreme care to prevent pollution of the ground water by these actions.

7. Measurement and payment

Method 1—Items of work listed in the bid schedule for removal of water, diverting surface water, and dewatering construction sites and borrow areas are paid for at the contract lump sum prices. Such payment will constitute full compensation for all labor, equipment, tools, and all other items necessary and incidental to the completion of the work.

Method 2—Items of work listed in the bid schedule for removal of water, diverting surface water, dewatering construction sites, and dewatering borrow areas are paid for at the contract lump sum prices. Such payment will constitute full compensation for furnishing, installing, operating, and maintaining the necessary trenches, drains, sumps, pumps, and piping and for all labor, equipment, tools, and all other items necessary and incidental to the completion of the work. The exception is that additional payment for pumping to dewater borrow areas and the removal of water will be made as described in the following paragraph.

If pumping to dewater borrow areas is a contract bid item, payment is made at the contract unit price, which shall be the price per 1,000 gallons shown in the bid schedule. Such payment will constitute full compensation for pumping only. Compensation for equipment and preparation and for other costs associated with pumping is included in the lump sum payment for removal of water or the lump sum payment for dewatering the borrow areas. Payment is made only for pumping that is necessary to dewater borrow areas that cannot be effectively drained by gravity or that must have the water table lowered to be usable as a suitable borrow source. Pumping for other purposes will not

be included for payment under this item.

All Methods—The following provisions apply to all methods of measurement and payment. Compensation for any item of work described in the contract, but not listed in the bid schedule is included in the payment for the contract line item to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in section 8 of this specification.

8. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details therefore are:

Subsidiary Item - Removal of Water

1. This item shall consist of all dewatering required for the work under this contract.
2. A written dewatering plan from the Contractor is required at the preconstruction meeting and shall be provided to the COR.
3. Method of Payment
Section 6 Measurement and Payment does not apply. No separate payment will be made for Removal of Water. Compensation for this item will be included in the payment for Bid Item 2 – Channel Clearing and Shaping.

Construction Specification 13—Piling

1. Scope

The work consists of furnishing and installing the specified kinds and types of piles at the locations shown on the drawings.

2. Material

Piles shall conform to the requirements of following material specifications as appropriate to the kinds of piles specified. For piles of materials other than those listed, the material requirements outlined in section 14 of this specification shall apply.

511—Steel Piles

512—Wood Piles

513—Precast Concrete Piles

514—Cast-in-Place Concrete Piles

3. Site preparation

All excavation within the area to be occupied by bearing piles shall be completed before the piles are driven.

4. Protection of pile heads

The heads of all piles shall be protected during driving by suitable caps, rings, heads, blocks, mandrels, and other devices.

The heads of timber piles shall be fitted into a steel head block or fitted with heavy steel or wrought iron rings or wire wrapping.

The heads of steel piles shall be cut square and fitted with a steel driving cap.

The heads of precast concrete piles and casings shall be fitted into cushion type drive caps having a rope or other suitable cushion next to the pile head and fitting into a casting that in turn supports a timber shock block.

Driving heads, mandrels, and other devices shall be provided by the contractor as needed for special types of piles and shall conform to the recommendations of the pile manufacturer.

5. Piles, general

The contractor shall notify the engineer before pile driving operation commences. Such notice shall be far enough in advance, a minimum of 24 hours, to provide the engineer adequate time to be present for the driving operations. Piles shall be driven only in the presence of the engineer or authorized representative.

The determination of piling order lengths shall be the contractor's responsibility unless otherwise specified.

Unless otherwise approved, piles shall be driven with steam, air, diesel powered hammers or a combination of hammers, or by vibration or water jets. Water jets may be

used only when specifically authorized by the engineer. Where jetting is authorized, the jets shall be withdrawn before the specified depth or bearing capacity is obtained and the piles shall be driven with the hammer to the final penetration.

When drop hammers are permitted, the height of drop shall not be more than 8 feet for concrete piles or 12 feet for steel and timber piles, unless otherwise specified.

The driving of piling with followers shall be allowed only when expressly approved by the engineer.

Piles shall not be driven within 20 feet of concrete less than 7 days after placement, including concrete placed in cast-in-place piles with or without pre-driven shells or casings.

The contractor shall not attempt to drive piles beyond the point of refusal, as indicated by excessive bouncing of the hammer or kicking of the pile.

6. Bearing piles

Bearing piles shall be driven to the position, line, and batter specified on the drawings. Each pile shall be driven continuously and without interruption to the specified depth or until the specified bearing capacity is obtained. Deviation from this procedure is permitted only when interruption of driving is caused by conditions that could not reasonably be anticipated.

When a diesel hammer is used, it shall be operated at full throttle when blows are counted for determination of bearing capacity except that throttle adjustments shall be made as necessary to prevent the non-striking parts of the hammer from rising from the pile on the ram upstroke.

7. Sheet piles

The piling shall be driven in a manner that ensures perfect interlocking throughout the entire length of each pile. The piles shall be held in proper alignment during driving by assembling frames or other suitable temporary guide structures. Temporary guide structures shall be removed when they have served their purpose.

Anytime the forward edge of the sheet pile wall is found to be out of correct alignment,

- a. The piling already assembled and partly driven shall be driven to the required depth.
- b. Taper piles shall then be driven to bring the forward edge into correct alignment before additional regular piling is assembled and driven. The maximum permissible taper in a single pile shall be 0.25 inch per foot of length.

8. Estimating bearing capacity

When load tests are not required, the bearing capacity of each pile shall be estimated using one of the following formulas, as appropriate:

Gravity hammers:

$$R = \frac{2WH}{S+1}$$

Single-acting steam or air hammers and diesel hammers having unrestricted rebound of the ram:

$$R = \frac{2WH}{S+0.1}$$

Double-acting steam or air hammers and diesel hammers having enclosed rams:

$$R = \frac{2H(W+AP)}{S+0.1} \quad \text{or} \quad R = \frac{2E}{S+0.1}$$

where:

- R = safe bearing capacity, in pounds
- W = weight of striking parts of hammer, in pounds
- H = height of fall, in feet
- A = area of piston, in square inches
- P = pressure of steam, air, or other gas exerted on the hammer piston or ram, in pounds per square inch
- E = the manufacturer's rating for foot-pounds of energy developed by double-acting steam or air hammers, or 90 percent of the average equivalent energy developed by diesel hammers having enclosed rams as evaluated by gauge and chart readings, in foot-pounds
- S = average penetration for the last 5 to 10 blows of a gravity hammer or the last 10 to 20 blows for steam, air, or diesel powered hammers, in inches per blow

These formulas are applicable when:

- The hammer has a free fall.
- The head of the pile is not crushed.
- The penetration is reasonably quick and uniform.
- There is no sensible bounce after the blow.
- A follower is not used.

Twice the height of the bounce shall be deducted from **H** to determine its value in the formula.

If case water jets are used in conjunction with the driving, these formulas are used to determine the bearing power from the results of driving after the jets have been removed.

9. Load tests

When load tests are specified, the test loads shall be applied gradually, without impact, and in a manner that no lateral forces are applied to the pile. Load testing shall not be started until 24 hours after driving of the test pile is completed unless otherwise specified in section 14 of this specification. Except as otherwise specified, load tests shall be performed according to the following procedures.

The total test load shall be twice the specified working load and shall be applied to the pile in increments equal to 25 percent of the working load. Settlement of the top of the pile shall be measured to an accuracy of 0.01 inch before and after the application of each load increment and at 2, 4, 8, 15, 30, and 60 minutes after, and then every 2 hours until the next load increment is applied. Additional load shall not be applied until the rate

of settlement is less than 0.01 inch in 1 hour.

The total test load shall remain on the pile for a minimum of 24 hours. Settlement shall be measured at 6-hour intervals during this period and at the end of the period, at least twice during removal of the load, and immediately after all of the test load is removed. The net settlement shall be measured about 24 hours after the total load has been removed.

If settlement continues in excess of 0.01 inch per hour under less than the total test load, no additional load shall be applied. However, the load that has been applied shall remain on the pile a minimum of 24 hours, and settlement measurements while the load is on the pile and during and after removal of the load shall be made as if it were the total test load.

10. Cutting off piles

The contractor shall cut the piles at the specified elevations. The length of pile cut off shall be sufficient to permit the removal of all damaged material. Steel shells or concrete casings for cast-in-place concrete piles shall be cut off at the specified elevation before being filled with concrete.

Steel bearing piles shall be cut off in clean, straight lines as shown on the drawings. Any irregularities shall be leveled off with deposits of weld metal or by grinding before placement of bearing caps.

Precast concrete piles and concrete casings shall be cut off in a manner that prevents damage to the rest of the pile or casing or to the projecting reinforcement required for connecting the piles to the structure.

Timber piles that are to be capped shall be accurately cut off so that true bearing is obtained on every pile without the use of shims.

11. Defective piles

Any pile damaged in driving, driven out of proper location, driven below the specified cutoff elevation, or inaccurately cut off shall be corrected by one of the following methods, as approved by the engineer:

- a. The defective pile shall be pulled and replaced or re-driven.
- b. A new pile shall be driven adjacent to the defective pile.
- c. The defective pile shall be spliced or built up or a sufficient part of the footing shall be extended to properly embed the pile.

Pile shells abandoned in place after driving shall be filled with concrete or sand-cement grout as appropriate to the conditions that are present.

All piles pushed up by the driving of adjacent piles or by any other cause shall be re-driven to final grade.

Any sheet pile ruptured in the interlock or otherwise damaged during driving shall be pulled and replaced.

12. Correcting surface heave

Any excess material resulting from displacement of earth by pile driving shall be removed. Materials disturbed by pile driving shall be conditioned and compacted to a minimum density equal to adjacent undisturbed material.

13. Measurement and payment

Method 1—For items of work for which specific unit prices are established in the contract, each type, kind, and length of pile driven in place is counted. Payment for furnishing and driving each type, kind, and length of pile is made at the contract unit price. Such payment will constitute full compensation for all labor, equipment, materials, and all other items necessary and incidental to the completion of the work.

Method 2—For items of work for which specific unit prices are established in the contract, each type, kind, and length of pile furnished, accepted, and stockpiled in good condition at the site of the work is counted. Payment for furnishing each type, kind, and length of pile is made at the contract unit price. Payment for driving each type and kind of pile is made at the contract unit price. Such payment will constitute full compensation for all labor, equipment, materials, and all other items necessary and incidental to the completion of the work.

Method 3—For items of work for which specific unit prices are established in the contract, the length of each type and kind of pile driven is computed to the nearest foot as the difference between the measured length of pile before driving and measured length of pile cut off after driving. Payment for furnishing and driving each type and kind of pile is made at the contract unit price. Such payment will constitute full payment for all labor, equipment, materials, and other items necessary and incidental to the completion of the work.

Method 4—For items of work for which specific unit prices are established in the contract, the area of sheet pile walls, acceptably placed in accordance and within the neat lines shown on the drawings, is computed to the nearest square foot. Payment is made at the contract unit price for each type, kind, and weight of piling. Such payment will constitute full payment for all labor, equipment, materials, and other items necessary and incidental to the completion of the work.

All Methods—The following provisions apply to all methods of measurement and payment:

The measurement of the number of linear feet of piles (or number of piles) furnished and the number of piles driven shall include test and tension piles specified in the contract. Piles furnished and driven at the option of the contractor are not included. No payment is made for furnishing or driving pile, including test piles, to replace piles lost or damaged before the completion of the contract while in stockpile or during handling and driving.

When load tests are specified, payment for each test is made at the contract unit price per test. Such payment will constitute full compensation for all labor, equipment, materials, and all other items necessary and incidental to perform the test, except furnishing and driving piling.

When splices are specified, payment for each splice is made at the contract unit price. Such payment shall constitute full compensation for labor, equipment, materials, and all other items necessary and incidental to the completion of the work.

Compensation for any item of work described in the contract, but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in section 14 of this specification.

14. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details therefore are:

Bid Item 3 - Piling

1. This item shall consist of furnishing and installing the steel sheet piling and piling cap including all forces, material, and equipment necessary for performing the work required under this item.
2. Sheet piles shall be equivalent to SKZ 20 cold formed steel sheet pile. Sheet piles shall be of minimum Grade 50 steel (50 ksi yield strength). Sheet piles shall have a minimum section modulus of 31.69 in³/ft. Sheet piles shall be of minimum 28.5 feet in length. All Sheet Piles shall be driven to the depth shown and described in the drawings. Bedrock is not anticipated to be encountered.
3. The steel piling cap shall be 8" x 8" x 1/2" steel angle stock and welded to each piece of sheet piling face with a standard 6-inch weld as shown and described in the drawings. No separate payment will be made for material and installation of the cap. Payment is to be included in the unit price of the Bid Item 3, Sheet Piling.
4. Alignment of the piling wall will be staked in the field by NRCS. The road-side face of the sheet pile wall shall be offset from the baseline as per the drawings.
5. OSHA approved handling devices or slings shall be used to set the sheet piling.
6. Construction equipment shall not be operated within 5 feet of the road-side face of the sheet pile wall during construction.
7. Holes for the 4" Sch. 40 PVC outlet pipe shall be cut using a hole saw, torch, or an alternative method approved by the COR. Cut holes shall not damage or impeded the function of the sheet pile interlocks. The hole shall be circular and have no larger than 0.5" gap from the outside diameter of the pipe. All hole edges shall be deburred prior to the installation of the PVC outlet pipe. Gaps between the PVC outlet pipe and the sheet pile hole shall be sealed with exterior silicone caulk.
8. Measurement and payment shall be by Method 3.

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Construction Specification 21—Excavation

1. Scope

The work shall consist of the excavation required by the drawings and specifications and disposal of the excavated materials.

2. Classification

Excavation is classified as common excavation, rock excavation, or unclassified excavation in accordance with the following definitions.

Common excavation is defined as the excavation of all materials that can be excavated, transported, and unloaded using heavy ripping equipment and wheel tractor-scrapers with pusher tractors or that can be excavated and dumped into place or loaded onto hauling equipment by excavators having a rated capacity of one cubic yard or larger and equipped with attachments (shovel, bucket, backhoe, dragline, or clam shell) appropriate to the material type, character, and nature of the materials.

Rock excavation is defined as the excavation of all hard, compacted, or cemented materials that require blasting or the use of ripping and excavating equipment larger than defined for common excavation. The excavation and removal of isolated boulders or rock fragments larger than 1 cubic yard encountered in materials otherwise conforming to the definition of common excavation shall be classified as rock excavation. The presence of isolated boulders or rock fragments larger than 1 cubic yard is not in itself sufficient cause to change the classification of the surrounding material.

For the purpose of these classifications, the following definitions shall apply:

Heavy ripping equipment is a rear-mounted, heavy duty, single-tooth, ripping attachment mounted on a track type tractor having a power rating of at least 250 flywheel horsepower unless otherwise specified in section 10.

Wheel tractor-scraper is a self-loading (not elevating) and unloading scraper having a struck bowl capacity of at least 12 cubic yards.

Pusher tractor is a track type tractor having a power rating of at least 250 flywheel horsepower equipped with appropriate attachments.

Unclassified excavation is defined as the excavation of all materials encountered, including rock materials, regardless of their nature or the manner in which they are removed.

3. Blasting

The transportation, handling, storage, and use of dynamite and other explosives shall be directed and supervised by a person(s) of proven experience and ability who is authorized and qualified to conduct blasting operations.

Blasting shall be done in a manner as to prevent damage to the work or unnecessary fracturing of the underlying rock materials and shall conform to any special requirements in section 10 of this specification. When specified in section 10, the contractor shall

furnish the engineer, in writing, a blasting plan before blasting operations begin.

4. Use of excavated material

Method 1—To the extent they are needed, all suitable material from the specified excavations shall be used in the construction of required permanent earthfill or rockfill. The suitability of material for specific purposes is determined by the engineer. The contractor shall not waste or otherwise dispose of suitable excavated material.

Method 2—Suitable material from the specified excavations may be used in the construction of required earthfill or rockfill. The suitability of material for specific purposes is determined by the engineer.

5. Disposal of waste materials

Method 1—All surplus or unsuitable excavated materials are designated as waste and shall be disposed of at the locations shown on the drawings.

Method 2—All surplus or unsuitable excavated materials are designated as waste and shall be disposed of by the contractor at sites of his own choosing away from the site of the work. The disposal shall be in an environmentally acceptable manner that does not violate local rules and regulations.

6. Excavation limits

Excavations shall comply with OSHA Construction Industry Standards (29CFR Part 1926) Subpart P, Excavations, Trenching, and Shoring. All excavations shall be completed and maintained in a safe and stable condition throughout the total construction phase. Structure and trench excavations shall be completed to the specified elevations and to the length and width required to safely install, adjust, and remove any forms, bracing, or supports necessary for the installation of the work. Excavations outside the lines and limits shown on the drawings or specified herein required to meet safety requirements shall be the responsibility of the contractor in constructing and maintaining a safe and stable excavation.

7. Borrow excavation

When the quantities of suitable material obtained from specified excavations are insufficient to construct the specified earthfills and earth backfills, additional material shall be obtained from the designated borrow areas. The extent and depth of borrow pits within the limits of the designated borrow areas shall be as specified in section 10 or as approved by the engineer.

Borrow pits shall be excavated and finally dressed to blend with the existing topography and sloped to prevent ponding and to provide drainage.

8. Overexcavation

Excavation in rock beyond the specified lines and grades shall be corrected by filling the resulting voids with portland cement concrete made of materials and mix proportions approved by the engineer. Concrete that will be exposed to the atmosphere when construction is completed shall meet the requirements of concrete selected for use under Construction Specification 31, Concrete for Major Structures, or 32, Structure Concrete, as appropriate.

Concrete that will be permanently covered shall contain not less than five bags of cement per cubic yard. The concrete shall be placed and cured as specified by the engineer.

Excavation in earth beyond the specified lines and grades shall be corrected by filling the resulting voids with approved, compacted earthfill. The exception to this is that if the earth is to become the subgrade for riprap, rockfill, sand or gravel bedding, or drainfill, the voids may be filled with material conforming to the specifications for the riprap, rockfill, bedding, or drainfill. Before correcting an overexcavation condition, the contractor shall review the planned corrective action with the engineer and obtain approval of the corrective measures.

9. Measurement and payment

For items of work for which specific unit prices are established in the contract, the volume of each type and class of excavation within the specified pay limits is measured and computed to the nearest cubic yard by the method of average cross-sectional end areas or by methods outlined in section 10 of this specification. Regardless of quantities excavated, the measurement for payment is made to the specified pay limits except that excavation outside the specified lines and grades directed by the engineer to remove unsuitable material is included. Excavation required because unsuitable conditions result from the contractor's improper construction operations, as determined by the engineer, is not included for measurement and payment.

Method 1—The pay limits shall be as designated on the drawings.

Method 2—The pay limits shall be defined as follows:

- a. The upper limit shall be the original ground surface as it existed before the start of construction operations except that where excavation is performed within areas designated for previous excavation or earthfill, the upper limit shall be the modified ground surface resulting from the specified previous excavation or earthfill.
- b. The lower and lateral limits shall be the neat lines and grades shown on the drawings.

Method 3—The pay limits shall be defined as follows:

- a. The upper limit shall be the original ground surface as it existed before the start of construction operations except that where excavation is performed within areas designated for previous excavation or earthfill, the upper limit shall be the modified ground surface resulting from the specified previous excavation or earthfill.
- b. The lower and lateral limits shall be the true surface of the completed excavation as directed by the engineer.

Method 4—The pay limits shall be defined as follows:

- a. The upper limit shall be the original ground surface as it existed before the start of construction operations except that where excavation is performed within areas designated for previous excavation or earthfill, the upper limit shall be the modified ground surface resulting from the specified previous excavation or earthfill.

- b. The lower limit shall be at the bottom surface of the proposed structure.
- c. The lateral limits shall be 18 inches outside of the outside surface of the proposed structure or shall be vertical planes 18 inches outside of and parallel to the footings, whichever gives the larger pay quantity, except as provided in d below.
- d. For trapezoidal channel linings or similar structures that are to be supported upon the sides of the excavation without intervening forms, the lateral limits shall be at the underside of the proposed lining or structure.
- e. For the purposes of the definitions in b, c, and d, above, any specified bedding or drainfill directly beneath or beside the structure will be considered to be a part of the structure.

All methods—The following provisions apply to all methods of measurement and payment.

Payment for each type and class of excavation is made at the contract unit price for that type and class of excavation. Such payment will constitute full compensation for all labor, materials, equipment, and all other items necessary and incidental to the performance of the work except that extra payment for backfilling overexcavation will be made in accordance with the following provisions.

Payment for backfilling overexcavation, as specified in section 8 of this specification, is made only if the excavation outside specified lines and grades is directed by the engineer to remove unsuitable material and if the unsuitable condition is not a result of the contractor's improper construction operations as determined by the engineer.

Compensation for any item of work described in the contract, but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in section 10 of this specification.

10. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details therefor are:

Subsidiary Item - Excavation (Unclassified)

1. This item shall consist of all excavation required to construct this project. This includes excavation under and behind the drainfill, rock riprap, and piling.
2. All excavation on this project is designated as "unclassified excavation" and shall consist of all materials encountered regardless of their nature or the manner in which they are removed.
3. Use of excavated material shall be by Method 1.
4. Disposal of waste material shall be by Method 2.
5. Method of Payment
Section 9 Measurement and Payment does not apply. No separate payment will be made for Excavation. Compensation for this item will be included in the payment for Bid Item 2 - Channel Clearing and Shaping.

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Construction Specification 23—Earthfill

1. Scope

The work consists of the construction of earth embankments, other earthfills, and earth backfills required by the drawings and specifications.

Earthfill is composed of natural earth materials that can be placed and compacted by construction equipment operated in a conventional manner.

Earth backfill is composed of natural earth material placed and compacted in confined spaces or adjacent to structures (including pipes) by hand tamping, manually directed power tampers or vibrating plates, or their equivalent.

2. Material

All fill material shall be obtained from required excavations and designated borrow areas. The selection, blending, routing, and disposition of material in the various fills shall be subject to approval by the engineer.

Fill materials shall contain no frozen soil, sod, brush, roots, or other perishable material. Rock particles larger than the maximum size specified for each type of fill shall be removed prior to compaction of the fill.

The types of material used in the various fills shall be as listed and described in the specifications and drawings.

3. Foundation preparation

Foundations for earthfill shall be stripped to remove vegetation and other unsuitable material or shall be excavated as specified.

Except as otherwise specified, earth foundation surfaces shall be graded to remove surface irregularities and shall be scarified parallel to the axis of the fill or otherwise acceptably scored and loosened to a minimum depth of 2 inches. The moisture content of the loosened material shall be controlled as specified for the earthfill, and the surface material of the foundation shall be compacted and bonded with the first layer of earthfill as specified for subsequent layers of earthfill.

Earth abutment surfaces shall be free of loose, uncompacted earth in excess of 2 inches in depth normal to the slope and shall be at such a moisture content that the earthfill can be compacted against them to produce a good bond between the fill and the abutments.

Rock foundation and abutment surfaces shall be cleared of all loose material by hand or other effective means and shall be free of standing water when fill is placed upon them. Occasional rock outcrops in earth foundations for earthfill, except in dams and other structures designed to restrain the movement of water, shall not require special treatment if they do not interfere with compaction of the foundation and initial layers of the fill or the bond between the foundation and the fill.

Foundation and abutment surfaces shall be no steeper than one horizontal to one vertical unless otherwise specified. Test pits or other cavities shall be filled with compacted earthfill conforming to the specifications for the earthfill to be placed upon the foundation.

4. Placement

Earthfill shall not be placed until the required excavation and foundation preparation have been completed and the foundation has been inspected and approved by the engineer. Earthfill shall not be placed upon a frozen surface nor shall snow, ice, or frozen material be incorporated in the earthfill matrix.

Earthfill shall be placed in approximately horizontal layers. The thickness of each layer before compaction shall not exceed the maximum thickness specified in section 10 or shown on the drawings. Materials placed by dumping in piles or windrows shall be spread uniformly to not more than the specified thickness before being compacted.

Hand compacted earth backfill shall be placed in layers whose thickness before compaction does not exceed the maximum thickness specified for layers of earth backfill compacted by manually directed power tampers.

Earth backfill shall be placed in a manner that prevents damage to the structures and allows the structures to assume the loads from the earth backfill gradually and uniformly. The height of the earth backfill adjacent to a structure shall be increased at approximately the same rate on all sides of the structure.

Earthfill and earth backfill in dams, levees, and other structures designed to restrain the movement of water shall be placed to meet the following additional requirements:

- (a) The distribution of materials throughout each zone shall be essentially uniform, and the earthfill shall be free from lenses, pockets, streaks, or layers of material differing substantially in texture, moisture content, or gradation from the surrounding material. Zone earthfills shall be constructed concurrently unless otherwise specified.
- (b) The surface of each layer shall be scarified parallel to the axis of the fill to a depth of not less than 2 inches before the next layer is placed.
- (c) The top surface of embankments shall be maintained approximately level during construction with two exceptions: A crown or cross-slope of about 2 percent shall be maintained to ensure effective drainage, or as otherwise specified for drainfill or sectional zones.
- (d) Dam embankments shall be constructed in continuous layers from abutment to abutment except where openings to facilitate construction or to allow the passage of streamflow during construction are specifically authorized in the contract.
- (e) Embankments built at different levels as described under (c) or (d) above shall be constructed so that the slope of the bonding surfaces between embankment in place and embankment to be placed is not steeper than 3 feet horizontal to 1 foot vertical. The bonding surface of the embankment in place shall be stripped of all material not meeting the requirements of this specification and shall be scarified, moistened, and recompactd when the new earthfill is placed against it. This ensures a good bond with the new earthfill and obtains the specified moisture content and density at the contact of the in-place and new earthfills.

5. Control of moisture content

During placement and compaction of earthfill and earth backfill, the moisture content of the material being placed shall be maintained within the specified range.

The application of water to the earthfill material shall be accomplished at the borrow areas insofar as practicable. Water may be applied by sprinkling the material after placement on the earthfill, if necessary. Uniform moisture distribution shall be obtained by disking.

Material that is too wet when deposited on the earthfill shall either be removed or be dried to the specified moisture content prior to compaction.

If the top surface of the preceding layer of compacted earthfill or a foundation or abutment surface in the zone of contact with the earthfill becomes too dry to permit suitable bond, it shall either be removed or scarified and moistened by sprinkling to an acceptable moisture content before placement of the next layer of earthfill.

6. Compaction

Earthfill—Earthfill shall be compacted according to the following requirements for the class of compaction specified:

Class A compaction—Each layer of earthfill shall be compacted as necessary to provide the density of the earthfill matrix not less than the minimum density specified in Section 10 or identified on the drawings. The earthfill matrix is defined as the portion of the earthfill material finer than the maximum particle size allowed in the reference compaction test method specified (ASTM D698 or ASTM D1557).

Class B compaction—Each layer of earthfill shall be compacted to a mass density not less than the minimum density specified.

Class C compaction—Each layer of earthfill shall be compacted by the specified number of passes of the type and weight of roller or other equipment specified or by an approved equivalent method. Each pass shall consist of at least one passage of the roller wheel or drum over the entire surface of the layer.

Earth backfill—Earth backfill adjacent to structures shall be compacted to a density equivalent to that of the surrounding in-place earth material or adjacent required earthfill or earth backfill. Compaction shall be accomplished by hand tamping or manually directed power tampers, plate vibrators, walk-behind, miniature, or self-propelled rollers. Unless otherwise specified heavy equipment including backhoe mounted power tampers or vibrating compactors and manually directed vibrating rollers shall not be operated within 3 feet of any structure. Towed or self-propelled vibrating rollers shall not be operated within 5 feet of any structure. Compaction by means of drop weights operating from a crane or hoist is not permitted.

The passage of heavy equipment will not be allowed:

- Over cast-in-place conduits within 14-days after placement of the concrete
- Over cradled or bedded precast conduits within 7 days after placement of the concrete cradle or bedding
- Over any type of conduit until the backfill has been placed above the top surface of the structure to a height equal to one-half the clear span width of the structure or pipe or 3 feet, whichever is greater, except as may be specified in section 10.

Compacting of earth backfill adjacent to structures shall not be started until the concrete

has attained the strength specified in section 10 for this purpose. The strength is determined by compression testing of test cylinders cast by the contractor's quality control personnel for this purpose and cured at the work site in the manner specified in ASTM C 31 for determining when a structure may be put into service.

When the required strength of the concrete is not specified as described above, compaction of earth backfill adjacent to structures shall not be started until the following time intervals have elapsed after placement of the concrete.

Structure interval	Time (days)
Vertical or near-vertical walls with earth loading on one side only	14
Walls backfilled on both sides simultaneously	7
Conduits and spillway risers, cast-in-place (with inside forms in place)	7
Conduits and spillway risers, cast-in-place (inside forms removed)	14
Conduits, pre-cast, cradled	2
Conduits, pre-cast, bedded	1
Cantilever outlet bents (backfilled both sides simultaneously)	3

7. Reworking or removal and replacement of defective earthfill

Earthfill placed at densities lower than the specified minimum density or at moisture contents outside the specified acceptable range of moisture content or otherwise not conforming to the requirements of the specifications shall be reworked to meet the requirements or removed and replaced by acceptable earthfill. The replacement earthfill and the foundation, abutment, and earthfill surfaces upon which it is placed shall conform to all requirements of this specification for foundation preparation, approval, placement, moisture control, and compaction.

8. Testing

During the course of the work, the contractor shall perform quality control tests, as applicable, to identify earthfill and earth backfill materials; determine the reference maximum density and optimum moisture content; and document that the moisture content of material at the time of compaction and the density of earthfill and earth backfill in place conform to the requirements of this specification.

Determining Reference Maximum Density and Optimum Moisture Content—For Class A compaction, the reference maximum density and optimum moisture content shall be determined in accordance with the compaction test and method specified on the drawings or in section 10.

Documenting Specification Conformance—In-place densities of earthfill and earth backfill requiring Class A compaction shall be measured in accordance with ASTM D1556, D2167, D2937, or D6938. Moisture contents of earthfill and earth backfill at the time of compaction shall be measured in accordance with ASTM D2216, D4643, or D6938. Values of moisture content determined by ASTM D2216 are considered the true value of the soil moisture. Values of moisture content determined by ASTM D4643 or D6938 shall be verified by comparison to values obtained by ASTM D2216. Values of in-place density and moisture content determined by these tests shall be compared to the minimum density and moisture content range specified on the drawings or in section 10.

Correction for Oversize Particles—If the materials to be used for earthfill or earth backfill contain more than 5 percent by dry weight of oversize rock particles (particles larger than those allowed in the specified compaction test and method), corrections for oversize particles shall be made using the appropriate procedures explained in ASTM D4718.

9. Measurement and payment

For items of work for which specific unit prices are established in the contract, the volume of each type and compaction class of earthfill and earth backfill within the specified zone boundaries and pay limits is measured and computed to the nearest cubic yard by the method of average cross-sectional end areas. Unless otherwise specified in section 10, no deduction in volume is made for embedded items, such as, but not limited to, conduits, inlet structures, outlet structures, embankment drains, sand diaphragm and outlet, and their appurtenances.

The pay limits shall be as defined below, with the further provision that earthfill required to fill voids resulting from overexcavation of the foundation, outside the specified lines and grades, will be included in the measurement for payment only under the following conditions:

- Where such overexcavation is directed by the engineer to remove unsuitable material, and
- Where the unsuitable condition is not a result of the contractor's improper construction operations as determined by the engineer.

Earthfill beyond the specified lines and grades to backfill excavation required for compliance with OSHA requirements will be considered subsidiary to the earthfill bid item(s).

Method 1—The pay limits shall be as designated on the drawings.

Method 2—The pay limits shall be the measured surface of the foundation when approved for placement of the earthfill and the specified neat lines of the earthfill surface.

Method 3—The pay limits shall be the measured surface of the foundation when approved for placement of the earthfill and the measured surface of the completed earthfill.

Method 4—The pay limits shall be the specified pay limits for excavation and the specified neat lines of the earthfill surface.

Method 5—The pay limits shall be the specified pay limits for excavation and the measured surface of the completed earthfill.

Method 6—Payment for each type and compaction class of earthfill and earth backfill is made at the contract unit price for that type and compaction class of earthfill. Such payment will constitute full compensation for all labor, material, equipment, and all other items necessary and incidental to the performance of the work.

Method 7—Payment for each type and compaction class of earthfill and earth backfill is made at the contract unit price for that type and compaction class of earthfill. Such payment will constitute full compensation for all labor, material, equipment, and all other items necessary and incidental to the performance of the work except furnishing, transporting, and applying water to the foundation and earthfill material. Water applied to the foundation and earthfill material is measured and payment made as specified in Construction Specification 10.

All methods—The following provisions apply to all methods of measurement and payment. Compensation for any item of work described in the contract, but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in section 10 of this specification.

10. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details therefor are:

Subsidiary Item - Earthfill (Class C)

1. This item shall consist of the placement of the Class C Earthfill as shown on the drawings.
2. Earthfill sourced from an off site location shall be reviewed and approved by the COR prior to placement.
3. Earthfill shall be placed with a maximum lift thickness of 12" prior to compaction. Earthfill lifts shall be placed and compacted horizontal.
4. Class C Compaction applies. The entire surface of each lift shall be traversed with 2 passes of the placing and spreading equipment.
5. The moisture content of the fill material shall be suitable for compaction, such that a hand molded ball will maintain its shape.
6. The top 12" of material shall be soil capable of supporting grass. The amount of rock present in the top 12" shall meet the requirements of Specification 6 Seeding, Sprigging, and Mulching.
7. Measurement and Payment
Section 9 Measurement and Payment does not apply. No separate payment will be made for Earthfill Class C. Compensation for this item will be included in Bid Item 2 - Channel Clearing and Shaping.

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Construction Specification 24—Drainfill

1. Scope

The work consists of furnishing, placing, and compacting drainfill required in the construction of structure drainage systems.

2. Material

Method 1—Drainfill material shall conform to the requirements of Material Specification 521, Aggregates for Drainfill and Filters. A minimum of 30 days before delivery of materials to the site, the contractor shall inform the engineer in writing of the source(s) from which drainfill material will be obtained. The contractor shall provide the engineer free access to the source(s) for the purpose of obtaining samples for testing.

Method 2—Drainfill material shall be sand, gravel, or crushed stone, or mixtures thereof, obtained from the specified sources. The material shall be selected as necessary to avoid the inclusion of organic matter, clay balls, excessive fine particles, or other substances that would interfere with their free-draining properties.

3. Base preparation

Foundation surface and trenches shall be clean and free of organic matter, loose soil, foreign substance, and standing water when the drainfill is placed. Earth surfaces upon or against which drainfill will be placed shall not be scarified.

4. Placement

Drainfill shall not be placed until the subgrade has been inspected and approved by the engineer. Drainfill shall not be placed over or around pipe or drain tile until the installation of the pipe or tile has been inspected and approved.

Drainfill shall be placed uniformly in layers not to exceed 12 inches thick before compaction. When compaction is accomplished by manually controlled equipment, the layers shall not exceed 8 inches thick. The material shall be placed to avoid segregation of particle sizes and to ensure the continuity and integrity of all zones. No foreign material shall be allowed to become intermixed with or otherwise contaminate the drainfill.

Traffic shall not be permitted to cross over drains at random. Equipment cross-overs shall be maintained, and the number and location of such crossovers shall be established and approved before the beginning of drainfill placement. Each crossover shall be cleaned of all contaminating material and shall be inspected and approved by the engineer before the placement of additional drainfill material.

Any damage to the foundation surface or the trench sides or bottom occurring during placement of drainfill shall be repaired before drainfill placement is continued.

The upper surface of drainfill constructed concurrently with adjacent zones of earthfill shall be maintained at a minimum elevation of 1 foot above the upper surface of adjacent earthfill.

Drainfill over and/or around pipe or drain tile shall be placed to avoid any displacement in line or grade of the pipe or tile.

Drainfill shall not be placed adjacent to structures until the concrete has attained the strength specified in section 9 of this specification. The strength shall be determined by compression testing of concrete test cylinders cast and field cured at the project site in accordance with ASTM Method C 31 for determining when a structure may be placed into service.

When the required strength of the concrete is not specified as described above, placement of drainfill adjacent to concrete structures shall not be commenced until the following item intervals have elapsed following placement of the concrete:

Structure type	Time interval (days)
Vertical or near-vertical wall with earth loading on one side only (retaining walls and counterforts)	14
Walls backfilled on both sides simultaneously	7
Conduits and galleries, cast-in-place (with inside forms in place)	7
(inside forms removed)	14
Conduits, precast, cradled	2
Conduits, precast, bedded	1
Cantilever outlet bents backfilled on both sides simultaneously	3

5. Control of moisture

The moisture content of drainfill material shall be controlled as specified in section 9 of this specification. When additional water is required, it shall be applied in a manner to avoid excessive wetting to adjacent earthfill. Except as specified in section 9 of this specification, control of moisture content will not be required.

6. Compaction

Drainfill shall be compacted according to the following requirements for the class of compaction specified:

Class A compaction—For drainfill materials with more than 70 percent passing the 3/4 inch sieve, each layer of drainfill shall be compacted to a minimum dry density of not less than the density specified in section 9 of this specification as determined by ASTM D 698. For drainfill materials with 70 percent or less passing the 3/4 inch sieve, each layer of drainfill shall be compacted to a relative density of not less than 70 percent as determined by ASTM D 4254.

Class I compaction—Each layer of drainfill shall be compacted by a minimum of two passes over the entire surface with a steel-drum vibrating roller weighing at least 5 tons and exerting a vertical vibrating force of not less than 20,000 pounds at a minimum frequency of 1,200 times per minute, or by an approved equivalent method.

Class II compaction—Each layer of drainfill shall be compacted by one of the following methods or by an approved equivalent method. (A pass is defined as at least one complete coverage of the roller wheel, tire, or drum over the entire surface for each layer.)

- a. A minimum of two passes over the entire surface with a pneumatic-tired roller exerting a minimum pressure of 75 pounds per square inch.
- b. A minimum of four passes over the entire surface with the track of a crawler-type tractor weighing at least 20 tons.
- c. Controlled movement of the hauling equipment so that the entire surface is traversed by not less than one tread track of the loaded hauling equipment.

Class III compaction—No compaction will be required beyond that resulting from the placing and spreading operations.

When compaction other than Class III compaction is specified, material placed in trenches or other locations inaccessible to heavy equipment shall be compacted by manually controlled pneumatic or vibrating tampers as specified in section 9 of this specification.

Heavy equipment shall not be operated within 2 feet of any structure. Vibrating rollers shall not be operated within 5 feet of any structure. Compaction by means of drop weights operating from cranes, hoists, or similar equipment will not be permitted.

7. Testing

The contractor shall conduct such tests as necessary to verify that the drainfill material and the in-place drainfill meets the specification requirements.

The engineer shall be granted access to perform such tests as are required to verify that the drainfill materials and the drainfill in place meets the requirements of the specifications. These tests are not intended to provide the contractor with information needed to assure that the materials and workmanship meet the specification requirements. These verification tests will not relieve the contractor of the responsibility of performing required tests for that purpose.

8. Measurement and payment

Method 1—For items of work for which specific unit prices are established in the contract, the volume of drainfill within the neat lines shown on the drawings are measured and computed to the nearest cubic yard. Where the engineer directs placement of drainfill outside the neat lines to replace unsuitable foundation material, the volume of such drainfill is included. The volume included is only to the extent that the unsuitable condition is not a result of the contractor's improper construction operation in the determination of the engineer.

Payment for drainfill is made at the contract unit price for each type of drainfill, complete in place. Except as otherwise specified in section 9 of this specification, such payment will constitute full compensation for all labor, equipment, material, and all other items necessary and incidental to the performance of the work.

Method 2—For items of work for which specific unit prices are established in the contract, the quantity of drainfill placed within the specified limits is computed to the nearest 0.1 ton by actual weight. Where the engineer directs placement of drainfill outside the neat lines to replace unsuitable foundation material, the weight of such drainfill is included. The weight included is only to the extent that the unsuitable condition is not a result of the contractor's improper construction operation in the determination of the engineer.

Payment for drainfill is made at the contract unit price for each type of drainfill, complete in place. Except as otherwise specified in section 9 of this specification, such payment will constitute full compensation for all labor, equipment, material, and all other items necessary and incidental to the performance of the work.

Compensation for any item of work described in the contract, but not included in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in section 9 of this specification.

9. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details therefore are:

Bid Item 6 – Drainfill (AASHTO M43 No. 57 Crushed Limestone Aggregate)

1. This item shall consist of furnishing and placing drainfill, geotextile, and the 4 inch drain pipe behind the sheet pile wall as shown and described in the drawings.
2. Material shall be by Method 2, as stated in section 2.
3. Material shall be crushed limestone from any source.
4. Drainfill furnished under this item shall meet the gradation requirements for: No. 57 aggregate as specified in table 703.1 of the most recent edition of the State of Ohio Department of Transportation, Construction and Material Specifications.
5. Drainfill compaction along the road side of the sheet pile wall shall be completed with a manually controlled pneumatic or vibrating tamper. Drainfill placed around the drain pipe shall be hand compacted.
6. Measurement and Payment shall be by Method 2, as stated in section 8.
7. Items subsidiary to this bid item are:
 - a. Construction Specification 45 - Plastic Pipe - Subsidiary Item – 4 inch Drain Pipe Perforated
 - b. Construction Specification 95 - Geotextile - Subsidiary Item - Nonwoven Geotextile Fabric

Subsidiary Item – Drainfill Bedding for Riprap Transitions
(AASHTO M43 No. 57 Crushed Limestone Aggregate)

1. This item shall consist of furnishing and placing drainfill bedding for the ODOT Type C riprap transitions as shown and described in the drawings.
2. Material shall be by Method 2, as stated in section 2.
3. Material shall be crushed limestone from any source.
4. Drainfill furnished under this item shall meet the gradation requirements for: No. 57 aggregate as specified in table 703.1 of the most recent edition of the State of Ohio Department of Transportation, Construction and Material Specifications.
5. Compaction shall be Class III, as stated in section 6.
6. Section 8 Measurement and Payment does not apply. No separate payment will be made for Subsidiary Item - Drainfill Bedding for Riprap Transitions (AASHTO M43 No. 57 Crushed Limestone Aggregate). Compensation for this item will be included in the payments for Bid Item 4– Rock Riprap Transition ODOT Type C.

Subsidiary Item – Drainfill Bedding for Riprap Toe
(AASHTO M43 No. 57 Crushed Limestone Aggregate)

1. This item shall consist of furnishing and placing drainfill bedding for the ODOT Type C riprap toe as shown and described in the drawings.
2. Material shall be by Method 2, as stated in section 2.
3. Material shall be crushed limestone from any source.
4. Drainfill furnished under this item shall meet the gradation requirements for: No. 57 aggregate as specified in table 703.1 of the most recent edition of the State of Ohio Department of Transportation, Construction and Material Specifications.
5. Compaction shall be Class III, as stated in section 6.
6. Section 8 Measurement and Payment does not apply. No separate payment will be made for Subsidiary Item - Drainfill Bedding for Riprap Toe (AASHTO M43 No. 57 Crushed Limestone Aggregate). Compensation for this item will be included in the payments for Bid Item 5– Rock Riprap Toe ODOT Type C.

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Construction Specification 45 — Plastic Pipe

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. The work consists of furnishing and installing plastic pipe (except corrugated polyethylene pipe) and the necessary fittings and appurtenances as shown on the drawings or as specified herein.

B. Material

1. Pipe, fittings, and gaskets must conform to the requirements of Material Specification 547, Plastic Pipe, and as specified in Section N of this specification or as shown on the drawings.
2. Perforated pipe must conform to the requirements of Material Specification 547, Plastic Pipe, and as specified in Section N of this specification or as shown on the drawings.
3. Unless otherwise specified, concrete must conform to the requirements of Construction Specification 32, Structure Concrete, and Section H of this specification.
4. Unless otherwise specified, earth backfill must conform to the requirements of Construction Specification 23, Earthfill.
5. Unless otherwise specified, drainfill must conform to the requirements of Construction Specification 24, Drainfill.

C. Handling and Storage

1. Deliver pipe to the job site and handled by means that provide adequate support to the pipe and do not subject it to undue stresses or damage. When handling and placing plastic pipe, take care to prevent impact blows, abrasion damage, and gouging or cutting by metal edges, surfaces or rocks. The manufacturer's special handling requirements must be strictly observed. Special care must be taken to avoid impact when the pipe must be handled at a temperature of 40 degrees Fahrenheit or less.
2. Store pipe on a relatively flat surface so that the barrels are evenly supported. Unless the pipe is specifically manufactured to withstand exposure to ultraviolet radiation, cover it with an opaque material when stored outdoors for 15 days or longer.

D. Excavation

1. Excavate in accordance with Construction Specification 21, Excavation, and Section N of this specification or as shown on the drawings.
2. Excavate the pipe foundation a minimum of 4 inches lower than the pipe grade shown on the drawings or staked in the field whenever bedrock, boulders, cobbles, or other material that may cause pipe damage is encountered at planned pipe grade.

E. Laying the Pipe

1. Install plastic pipe conduits complete with fittings and other related appurtenances to the lines and grades shown on the drawings or specified in Section N of this specification. Install the pipe so that there is no reversal of grade between joints unless otherwise shown on the drawings. The pipe must not be dropped or dumped on the bedding or into the pipe trench. The ground surface near the pipe trench must be free of loose rocks and stones greater than 1 inch in diameter. This ensures that rock will not be displaced and impact the pipe.
2. Just before placement, inspect each pipe section to ensure that all foreign material is removed from inside the pipe. The pipe ends and the couplings must be free of foreign material when assembled. At the completion of a work shift, all open ends of the pipeline must be temporarily closed off using a suitable cover or plug.
3. Take care to prevent distortion and damage during hot or cold weather. During unusually hot weather (daytime high temperature of more than 90 °F), lightly backfill the pipe assembled in the trench or shade to keep it as near to ground temperature as possible until final backfill is placed. Backfill operations must be performed during daily construction periods when the ground temperature and the temperature of the pipe do not vary more than 40 degrees Fahrenheit.
4. Install perforated pipe with the perforations down and oriented symmetrically about the vertical centerline. Perforations must be clear of any obstructions on the inside and outside of the pipe when the pipe is approved by the engineer for backfill.
5. During installation, firmly and uniformly bed the pipe throughout its entire length to the depth and in the manner specified in Section N of this specification or as shown on the drawings. Place bell-holes in bedding material under bells, couplings, and other fittings to ensure the pipe is uniformly supported throughout its entire length. Blocking or mounding beneath the pipe to bring the pipe to final grade is not permitted.

F. Pipe Embedment

1. Earth bedding.
 - a. Firmly and uniformly place the pipe on compacted earthfill bedding or an in-place earth material bedding of ample bearing strength to support the pipe without noticeable settlement. The earth material on which the pipe is placed must have uniform density to prevent differential settlement.

- b. Unless otherwise specified, form a groove that closely conforms to the outside surface of the pipe in the bedding. The depth of the groove must be equal to or greater than 0.3 of the pipe diameter.
 - c. Compact earth bedding to a density not less than adjacent, undisturbed, in-place earth material or be compacted earth backfill. Earthfill material used for compacted earth bedding must be free of rocks or stones greater than 1 inch in diameter and earth clods greater than 2 inches in diameter. The pipe must be loaded sufficiently during the compaction of bedding under the haunches and around the sides of the pipe to prevent displacement from its final approved placement.
2. Sand, gravel, or crushed rock bedding. When sand, gravel, or crushed rock bedding is specified, firmly and uniformly place the pipe on the bedding material. Material for bedding must not exceed 1 inch in diameter. Unless otherwise specified in Section N of this specification or shown on the drawings, the coarse-grained bedding material must be carefully placed and compacted to a depth equal to or greater than 0.3 of the diameter of the pipe above the bottom of the pipe. The pipe must be loaded sufficiently during backfilling and compaction around the sides to prevent displacement of the pipe from its final approved placement.
 3. Pipe encased in drainfill. Firmly and uniformly place the pipe on bedding of specified drainfill, which must be placed and compacted as specified in Section N of this specification or as shown on the drawings to form a continuous, uniform support around the entire circumference of the pipe. The pipe must be loaded sufficiently during backfilling around the sides and during compaction to prevent displacement of the pipe.

G. Backfill

1. Initial backfill.
 - a. Unless otherwise specified, initial backfill to 6 inches above the top of the conduit is required. Earth haunching and initial backfill material must consist of soil material free of rocks, stones, or hard clods more than 1 inch in diameter. Coarse backfill material must be the specified sand, gravel, crushed rock, or drainfill material.
 - b. Initial backfill must be placed in two stages. In the first stage (haunching), backfill is placed to the pipe spring line (center of pipe). In the second stage, it is placed to 6 inches above the top of the pipe.
 - c. The first stage material must be worked carefully under the haunches of the pipe to provide continuous support throughout the entire pipe length. The haunching backfill material must be placed in layers that have a maximum thickness of about 6 inches and are compacted as specified in Section N of this specification or as shown on the drawings. During compaction operations, care must be taken to ensure that the tamping or vibratory equipment does not come in contact with the pipe, and the pipe is not deformed or displaced.
 - d. When pressure testing is not specified, cover the pipe with a minimum of 6 inches of backfill material as soon as possible following assembly of the pipe in the trench but not later than within the same day that placement has

occurred. When pressure testing is specified, place sufficient backfill material over the pipe to anchor the conduit against movement during pressure testing activities.

2. Final backfill.

- a. Final backfill consists of placing the remaining material required to complete the backfill from the top of the initial backfill to the ground surface, including mounding at the top of the trench. Final backfill material within 2 feet of the top of the pipe must be free of debris or rocks larger than 3 inches nominal diameter. Coarse backfill material must be the specified sand, gravel, crushed rock, or drainfill. Final backfill must be placed in approximately uniform, compacted layers. Final backfill compaction requirements must be as specified in Section N of this specification or as shown on the drawings.
- b. Vehicles or construction equipment must not be allowed to cross the pipe until the minimum earth cover and required density as specified in Section 14 of this specification have been obtained.

H. Pipe Encasement in Concrete

1. Concrete encasement must be carefully placed to form a continuous, uniform support around the entire circumference of the pipe as specified in Section N of this specification or as shown on the drawings. Pipes encased in concrete must be securely anchored to prevent movement of the pipe during concrete placement. A clear distance of 1.5 inch must be maintained between the pipe and the reinforcing steel.
2. The concrete for the encasement must conform to the requirements of Construction Specification 32, Structure Concrete, for Class 3000M concrete unless otherwise specified.

I. Joints

1. Unless otherwise specified in Section N of this specification or shown on the drawings, joints must be either bell-and-spigot types with elastomeric gaskets, coupling types, solvent cement bell and spigot, or jointed by butt heat fusion. When a lubricant is required to facilitate joint assembly, it must be a type having no deleterious effect on the gasket or pipe material.
2. Pipe joints must be watertight at the pressures specified except where unsealed joints are indicated.
3. Install pipe and join in accordance with the manufacturer's recommendations. Laying deflections and joint fitting or stab depths must be within the manufacturer's recommended tolerances.
4. When solvent cement joints are specified for PVC or ABS pipe and fittings, they must be made in accordance with the following ASTMs and the related appendix of each ASTM: D2855 for PVC pipe and fittings and D2235 for ABS pipe and fittings.

5. Flanged, banded, heat-fusion, or elastomeric-sealed mechanical joints must be used when joining polyethylene (PE) and high-density polyethylene (HDPE) pipe and fittings unless otherwise specified in Section N of this specification or as shown on the drawings.
6. Cut pipe ends square and debur to provide a uniform, smooth surface for the jointing process. Place reference marks on the spigot ends to assist in determining when proper seating depth has been achieved within the joint.

J. Fittings

1. Unless otherwise specified, steel fittings, valves, and bolted connections must be painted or coated as recommended by the manufacturer.
2. Fittings for non-pressure pipe must be of the same or similar material as the pipe and must provide the same durability, watertightness, and strength as the pipe unless otherwise specified.

K. Thrust Blocks and Anchors

1. When specified, install concrete thrust blocks and anchors as shown on the drawings or as specified in Section N of this specification.
2. The concrete for thrust blocks and anchors must conform to the requirements of Construction Specification 32, Structure Concrete, for Class 3000M concrete unless otherwise specified in Section N of this specification.
3. The thrust block cavity must be hand-dug into undisturbed soil or previously placed, compacted backfill. The cavity must be formed with soil or wood to hold the freshly placed concrete without displacement until an initial set has occurred.
4. When excavation beyond the designated trench widths and depths as shown on the drawings or specified in Section N of this specification occurs at locations where installation of concrete thrust blocks is required, the contractor must install an alternative thrust block provision. The concrete thrust block must have a thickness of one pipe diameter and a contact face area that must be formed against the pipe as shown on the drawings or as specified in Section N of this specification. Place backfill on all sides of the thrust block and to the sides of the excavation. It must be compacted as specified for initial backfill.

L. Pressure Testing

1. Method 1. Pressure testing of the completed conduit is not required.
2. Method 2. Test the conduit for leaks in the following manner:
 - a. Before pressure testing:
 - (1) Joints of the assembled pipeline must be allowed to cure as recommended by the manufacturer.
 - (2) Flush and clean the pipeline.
 - (3) Place all concrete anchors and thrust blocks and allowed to cure for a minimum of 3 days.

- (4) Earth backfill must be sufficient to anchor the conduit against movement during the pressure testing and must be compacted as specified in Section N of this specification or as shown on the drawings.
 - (5) The conduit must be braced, anchored, or both at each end to restrict all potential pipe movement.
 - (6) Plug the ends of the conduit. The upstream plug must have a standpipe installed vertically with a minimum diameter of 2 inches and must be equipped with a shutoff valve. All high points in the line must be vented to evacuate air pockets. The conduit and the standpipe must be slowly filled with water so no air is entrapped during the filling operation. After filling is complete, all vents must be closed.
 - b. During pressure testing, the water level in the standpipe must be continuously maintained at a minimum of 10 feet above the highest invert elevation of the conduit for no less than 2 hours unless otherwise specified in Section N of this specification or as shown on the drawings.
 - (1) The volume of water leakage in the 2-hour test period must be recorded. The maximum allowable leakage (L) in gallons per hour must not exceed 0.02 times the nominal pipe diameter (D) in inches for each 1,000 feet of pipeline, which is about 50 pipe joints ($L = 0.02 \times D$).
 - c. When observed leakage exceeds the allowable amount, leaks must be sealed by replacement of pipe and fittings as necessary. The conduit must be retested as described above. This procedure must be repeated until the conduit leakage does not exceed the allowable volume specified above.
 - (1) The contractor must be fully responsible for any and all work required to correct leakage exceeding the amount specified.
3. Method 3. Test the conduit for leaks in the following manner:
- a. Before pressure testing:
 - (1) Joints of the assembled pipeline must be allowed to cure as recommended by the manufacturer.
 - (2) Flush and clean the pipeline.
 - (3) All concrete anchor and thrust blocks must be in place and allowed to cure for at least 3 days.
 - (4) Earth backfill must be sufficient to anchor the conduit against movement during the pressure testing and compacted as specified in Section N of this specification or as shown on the drawings.
 - (5) The conduit must be braced or anchored at each end to prevent all potential pipe movement.
 - (6) Plug the ends of the conduit and attach a pressure gauge to the upstream and downstream ends. All high points along the pipeline must be vented to permit the complete removal of all air within the pipeline. Slowly fill the conduit with water so no air is entrapped during the filling operations.
 - b. The testing pressure specified in Section N of this specification must be continuously maintained at the upstream gauge for a minimum of 2 hours. The pressure at the downstream gauge must not exceed the pressure rating of the pipe.

- c. The volume of water leakage for the 2-hour test period must be recorded. Maximum allowable leakage must be in accordance with figure 45-1 or equation 45.1:

Figure 45-1 Allowable leakage for plastic pipe (gal/hr/1,000 feet, or 50 pipe joints)^{1/}

Nominal Pipe Size (in)	Test Pressure in the Pipeline (lb/in ²)			
	50	100	150	200
	-----Allowable Leakage -----			
4	.19	.27	.33	.38
6	.29	.41	.50	.57
8	.38	.54	.66	.76
10	.48	.68	.83	.96
12	.57	.81	.99	1.15
14	.67	.95	1.16	1.34
15	.72	1.02	1.25	1.44
16	.76	1.07	1.32	1.52
18	.86	1.22	1.49	1.72

1/ Computation basis

$$L = \frac{ND \sqrt{P}}{7,400}$$

Eq: 45.1

where:

L = allowable leakage in gallons per hour

N = number of joints (pipe and fittings)

D = nominal diameter of pipe in inches

P = test pressure in pounds per square inch

- d. When observed leakage exceeds the allowable volume, leaks must be sealed by replacement of pipe and fittings as necessary. The conduit must be retested as described in this section. The procedure must be repeated until the conduit leakage does not exceed the allowable amount specified above.
- e. The contractor must be fully responsible for any and all work required to correct leakage exceeding the amount specified.

M. Measurement and Payment

1. Method 1. For items of work for which specific unit prices are established in the contract, the quantity of each kind, size, and class of pipe is determined to the nearest foot by measurement of the laid length along the crown centerline of the conduit. Payment for each kind, size, and class of pipe is made at the contract unit price for that kind, size, and class. Such payment constitutes full compensation for furnishing, transporting, and installing the pipe including excavation, shoring, backfill, bedding, thrust blocks, and all fittings, appurtenances, and other items necessary and incidental to the completion of the work. Payment for appurtenances listed separately in the bid schedule is made at the contract prices for those items.

2. Method 2. For items of work for which specific unit prices are established in the contract, the quantity of each kind, size, and class of pipe is determined as the sum of the nominal laying lengths of the sections used. Payment for each kind, size, and class of pipe is made at the contract unit price for the kind, size, and class. Such payment constitutes full compensation for furnishing, transporting, and installing the pipe including excavation, shoring, backfill, bedding, thrust blocks, and all fittings, appurtenances, and other items necessary and incidental to the completion of the work. Payment for appurtenances listed separately in the bid schedule is made at the contract prices for those items.
3. Method 3. For items of work for which specific unit prices are established in the contract, the quantity of each kind, size, and class of pipe is determined to the nearest foot by measurement of the laid length along the crown centerline of the conduit. Payment for each kind, size, and class of pipe is made at the contract unit price for the kind, size, and class. Such payment constitutes full compensation for furnishing, transporting, and installing the pipe including shoring, all fittings, thrust blocks, appurtenances, and other items necessary and incidental to the completion of the work. Payment for appurtenances listed separately in the bid schedule is made at the contract prices for those items.
4. Method 4. For items of work for which specific unit prices are established in the contract, the quantity of each kind, size, and class of pipe is determined as the sum of the nominal laying lengths of the pipe sections used. Payment for each kind, size, and class of pipe is made at the contract unit price for that kind, size, and class. Such payment constitutes full compensation for furnishing, transporting, and installing the pipe including shoring, all fittings, thrust blocks, appurtenances, and other items necessary and incidental to the completion of the work. Payment for appurtenances listed separately in the bid schedule is made at the contract prices for those items.
5. Methods 3 and 4. Excavation, backfill, and bedding are paid separately under their respective bid items.
6. All Measurement and Payment Methods. Compensation for any items of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and items to which they are made subsidiary are identified in Section N of this specification.

N. Items of Work and Construction Details

Items of work to be performed in conformance with this specification and the construction details therefore are:

Subsidiary Item – 4 inch Drain Pipe Perforated

1. This item shall consist of furnishing and installing the 4 inch perforated drain pipe.
2. The 4 inch pipe shall meet the minimum requirements for AASHTO M252.
3. The 4 inch pipe shall be installed at the base of the sheet pile wall along the backfill sides per the drawings. The pipe shall be placed a minimum 0.4% grade as shown in the drawings.
4. The outlet ends of the drain pipe will be connected to a ten foot section of solid 4 inch Schedule 40 PVC pipe meeting ASTM D1785. The PVC pipe will be ran through the tieback piles and extend between 6 to 12 inches past the outside of the pile wall.
5. An animal guard shall be installed at the pipe outlet as shown per the drawings.
6. Section Measurement and Payment does not apply. No separate payment will be made for Subsidiary Item - 4 inch Drain Pipe Perforated. Compensation for this item will be included in the Bid Item 6 – Drainfill (AASHTO M43 No. 57 Crushed Limestone Aggregate).

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Construction Specification 61 - Rock Riprap

1. Scope

The work consists of the construction of rock riprap revetments and blankets, including filter or bedding where specified.

2. Material

Rock riprap must conform to the requirements of Material Specification 523, Rock for Riprap, or if so specified, must be obtained from designated sources. It must be free from dirt, clay, sand, rock fines, and other material not meeting the required gradation limits.

At least 30 days before rock is delivered from other than designated sources, the contractor must designate in writing the source from which rock material will be obtained and provide information satisfactory to the contracting officer that the material meets contract requirements. The contractor must provide the responsible engineer free access to the source for the purpose of obtaining samples for testing. The size and grading of the rock must be as specified in section 8.

Rock from approved sources must be excavated, selected, and processed to meet the specified quality and grading requirements at the time the rock is installed.

Based on a specific gravity of 2.65 (typical of limestone and dolomite) and assuming the individual rock is shaped midway between a sphere and a cube, typical size/weight relationships are:

Sieve size of rock	Approx. weight of rock	Weight of test pile
16 inches	300 pounds	6,000 pounds
11 inches	100 pounds	2,000 pounds
6 inches	15 pounds	300 pounds

When specified in section 8 or when it is necessary to verify the gradation of the rock riprap, a particle size analysis must be performed in accordance with ASTM D5519, Test Method A or B. The analysis must be performed at the work site on a test pile of representative rock. The mass of the test pile must be at least 20 times the mass of the largest rock in the pile. The results of the test are compared to the gradation required for the project. Test pile results that do not meet the construction specifications must be cause for the rock to be rejected. The test pile that meets contract requirements must be left on the job site as a sample for visual comparison. The test pile must be used as part of the last rock riprap to be placed.

Filter or bedding aggregates when required must conform to Material Specification 521, Aggregates for Drainfill and Filters, unless otherwise specified. Geotextiles must conform to Material Specification 592, Geotextile.

3. Subgrade preparation

The subgrade surface on which the rock riprap, filter, bedding, or geotextile is to be placed must be cut or filled and graded to the lines and grades shown on the drawings. When fill to subgrade lines is required, it must consist of approved material and must conform to the requirements of the specified class of earthfill.

Rock riprap, filter, bedding, or geotextile must not be placed until the foundation preparation is completed and the subgrade surface has been inspected and approved.

4. Equipment-placed rock riprap

The rock riprap must be placed by equipment on the surface and to the depth specified. It must be installed to the full course thickness in one operation and in such a manner as to avoid serious displacement of the underlying material. The rock for riprap must be delivered and placed in a manner that ensures the riprap in place is reasonably homogeneous with the larger rocks uniformly distributed and firmly in contact one to another with the smaller rocks and spalls filling the voids between the larger rocks. Some hand placing may be required to provide a neat and uniform surface.

Rock riprap must be placed in a manner to prevent damage to structures. Hand placing is required as necessary to prevent damage to any new and existing structures.

5. Hand placed rock riprap

The rock riprap must be placed by hand on the surface and to the depth specified. It must be securely bedded with the larger rocks firmly in contact one to another without bridging. Spaces between the larger rocks must be filled with smaller rocks and spalls. Smaller rocks must not be grouped as a substitute for larger rock. Flat slab rock must be laid on its vertical edge except where it is laid like paving stone and the thickness of the rock equals the specified depth of the riprap course.

6. Filter or bedding

When the contract specifies filter, bedding, or geotextile beneath the rock riprap, the designated material must be placed on the prepared subgrade surface as specified. Compaction of filter or bedding aggregate is not required, but the surface of such material must be finished reasonably smooth and free of mounds, dips, or windrows.

7. Measurement and payment

Method 1—For items of work for which specific unit prices are established in the contract, the quantity of each type of rock riprap placed within the specified limits is computed to the nearest ton by actual weight. The volume of each type of filter or

bedding aggregate is measured within the specified limits and computed to the nearest cubic yard by the method of average cross-sectional end areas. For each load of rock riprap placed as specified, the contractor must furnish to the responsible engineer a statement-of-delivery ticket showing the weight to the nearest 0.1 ton.

Payment is made at the contract unit price for each type of rock riprap, filter, or bedding. Such payment is considered full compensation for completion of the work.

Method 2—For items of work for which specific unit prices are established in the contract, the quantity of each type of rock riprap placed within the specified limits is computed to the nearest 0.1 ton by actual weight. The quantity of each type of filter or bedding aggregate delivered and placed within the specified limits is computed to the nearest 0.1 ton. For each load of rock riprap placed as specified, the contractor must furnish to the engineer a statement-of-delivery ticket showing the weight to the nearest 0.1 ton. For each load of filter or bedding aggregate, the contractor must furnish to the responsible engineer a statement-of-delivery ticket showing the weight to the nearest 0.1 ton.

Payment is made at the contract unit price for each type of rock riprap, filter, or bedding. Such payment is considered full compensation for completion of the work.

Method 3—For items of work for which specific unit prices are established by the contract, the volume of each type of rock riprap and filter or bedding aggregate is measured within the specified limits and computed to the nearest cubic yard by the method of average cross-sectional end areas.

Payment is made at the contract unit price for each type of rock riprap, filter, or bedding. Such payment is considered full compensation for completion of the work.

Method 4—For items of work for which specific unit prices are established by the contract, the volume of each type of rock riprap, including filter and bedding aggregate, is measured within the specified limits and computed to the nearest cubic yard by the method of average cross-sectional end areas.

Payment is made at the contract unit price for each type of rock riprap, including filter and bedding. Such payment is considered full compensation for completion of the work.

Method 5—For items of work for which specific unit prices are established by the contract, the quantity of each type of rock riprap placed within the specified limits is computed to the nearest ton by actual weight. For each load of rock for riprap placed as specified, the contractor must furnish to the responsible engineer a statement-of-delivery ticket showing the weight to the nearest 0.1 ton.

Payment is made at the contract unit price for each type of rock riprap, and includes compensation for any aggregate or geotextile installed as specified for filter or bedding. Such payment is considered full compensation for completion of the work.

Method 6—For items of work for which specific unit prices are established by the contract, the volume of each type of rock riprap is measured within the specified limits

and computed to the nearest cubic yard by the method of average cross-sectional end areas.

Payment is made at the contract unit price for each type of rock riprap and includes compensation for any aggregate or geotextile installed as specified for filter or bedding. Such payment is considered full compensation for completion of the work.

All methods—The following provision applies to all methods of measurement and payment. Compensation for any item of work described in the contract, but not listed in the bid schedule, is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in section 8.

No separate payment is made for testing the gradation of the test pile. Compensation for testing is included in the appropriate bid item for riprap.

8. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details therefore are:

Bid Item 4– Rock Riprap Transition ODOT Type C

1. This item shall consist of furnishing and placing the equipment-placed Rock Riprap as shown and described in the drawings.
2. Rock Riprap furnished under this item shall meet the gradation requirements for: Type C, as described in section 703.19 of the most recent edition of the ODOT Construction and Material Specification Manual:
 - At least 85% of the material by weight shall be larger than a 6-inch but less than an 18-inch square opening.
 - At least 50% of the material by weight shall be larger than a 12-inch square opening.
 - The material smaller than a 6-inch square opening shall consist predominantly of rock spalls and rock fines and shall be free of soil.
3. Approved sources for the riprap are all suppliers specified in the most current Ohio Department of Transportation list of Certified Suppliers.
<https://cmsportal.dot.state.oh.us/>
(Select “Contractor Reports”, and then select “Certified Aggregate”)
4. Trucks delivering ODOT Type C riprap to the site shall not line their beds with smaller aggregate which contaminates the riprap.
5. The finished surface of the Rock Riprap shall not project more than 3 inches above the adjacent grades or the planned surface.
6. Method of Payment shall be by Method 5, as stated in section 7.
7. Items subsidiary to this bid item are:
 - Subsidiary Item - Drainfill Bedding for Riprap Transition (AASHTO M43 No. 57 Crushed Limestone Aggregate)

Bid Item 5– Rock Riprap Toe ODOT Type C

1. This item shall consist of furnishing and placing the equipment-placed Rock Riprap as shown and described in the drawings.
2. Rock Riprap furnished under this item shall meet the gradation requirements for: Type C, as described in section 703.19 of the most recent edition of the ODOT Construction and Material Specification Manual:
 - At least 85% of the material by weight shall be larger than a 6-inch but less than an 18-inch square opening.
 - At least 50% of the material by weight shall be larger than a 12-inch square opening.
 - The material smaller than a 6-inch square opening shall consist predominantly of rock spalls and rock fines and shall be free of soil.
3. Approved sources for the riprap are all suppliers specified in the most current Ohio Department of Transportation list of Certified Suppliers.
<https://cmsportal.dot.state.oh.us/>
(Select “Contractor Reports”, and then select “Certified Aggregate”)
4. Trucks delivering ODOT Type C riprap to the site shall not line their beds with smaller aggregate which contaminates the riprap.
5. The finished surface of the Rock Riprap shall not project more than 3 inches above the adjacent grades or the planned surface.
6. Method of Payment shall be by Method 5, as stated in section 7.
7. Items subsidiary to this bid item are:
 - Subsidiary Item - Drainfill Bedding for Riprap Toe (AASHTO M43 No. 57 Crushed Limestone Aggregate)

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Construction Specification 94—Contractor Quality Control

1. Scope

The work consists of developing, implementing, and maintaining a quality control system to ensure that the specified quality is achieved for all materials and work performed.

2. Equipment and materials

Equipment and material used for quality control shall be of the quality and condition required to meet the test specifications cited in the contract. Testing equipment shall be properly adjusted and calibrated at the start of operations and the calibration maintained at the frequency specified. Records of equipment calibration tests shall be available to the engineer at all times. Equipment shall be operated and maintained by qualified operators as prescribed in the manufacturer's operating instructions, the references specified, and as specified in section 10 of this specification. All equipment and materials used in performing quality control testing shall be as prescribed by the test standards referenced in the contract or in section 10.

All equipment and materials shall be handled and operated in a safe and proper manner and shall comply with all applicable regulations pertaining to their use, operation, handling, storage, and transportation.

3. Quality control system

Method 1—The contractor shall develop, implement, and maintain a system of quality control to provide the specified material testing and verification of material quality before use. The system activities shall include procedures to verify adequacy of completed work, initiate corrective action to be taken, and document the final results. The identification of the quality control personnel and their duties and authorities shall be submitted to the contracting officer in writing within 15 calendar days after notice of award.

Method 2—The contractor shall develop, implement, and maintain a system adequate to achieve the specified quality of all work performed, material incorporated, and equipment furnished before use. The system established shall be documented in a written plan developed by the contractor and approved by the contracting officer. The system activities shall include the material testing and inspection needed to verify the adequacy of completed work and procedures to be followed when corrective action is required. Daily records to substantiate the conduct of the system shall be maintained by the contractor. The quality control plan shall cover all aspects of quality control and shall address, as a minimum, all specified testing and inspection requirements. The plan provided shall be consistent with the planned performance in the contractor's approved construction schedule. The plan shall identify the contractor's onsite quality control manager and provide an organizational listing of all quality control personnel and their specific duties. The written plan shall be submitted to the contracting officer within 15 calendar days after notice of award. The contractor shall not proceed with any construction activity that requires inspection until the written plan is approved by the contracting officer.

All methods—The quality control system shall include, but not be limited to, a rigorous examination of construction material, processes, and operation, including testing of material and examination of manufacturer's certifications as required, to verify that work meets contract requirements and is performed in a competent manner.

4. Quality control personnel

Method 1—Quality control activities shall be accomplished by competent personnel. A competent person is: One who is experienced and capable of identifying, evaluating, and documenting that materials and processes being used will result in work that complies with the contract; and, who has authority to take prompt action to remove, replace, or correct such work or products not in compliance. Off-site testing laboratories shall be certified or inspected by a nationally recognized entity. The Contractor shall submit to the Contracting Officer, for approval, laboratory certification or inspection information. The Contractor shall submit to the Contracting Officer, for approval, the names, qualifications, authorities, certifications, and availability of the competent personnel who will perform the quality control activities.

Method 2—Quality control activities shall be accomplished by competent personnel who are separate and apart from line supervision and who report directly to management. A competent person is one who is experienced and capable of identifying, evaluating, and documenting that material and processes being used will result in work that complies with the contract, and who has authorization to take prompt action to remove, replace, or correct such work or products not in compliance. Offsite testing laboratories shall be certified or inspected by a nationally recognized entity. The Contractor shall submit to the Contracting Officer, for approval, laboratory certification or inspection information. The contractor shall submit to the contracting officer, for approval, the names, qualifications, authorities, certifications, and availability of the competent personnel who will perform the quality control activities.

5. Post-award conference

The contractor shall meet with the contracting officer before any work begins and discuss the contractor's quality control system. The contracting officer and the contractor shall develop a mutual understanding regarding the quality control system, including procedures for correcting quality control issues.

6. Records

The contractor's quality control records shall document both acceptable and deficient features of the work and corrective actions taken. All records shall be on forms approved by the contracting officer, be legible, and be dated and signed by the competent person creating the record.

Unless otherwise specified in section 10 of this specification, records shall include:

- a. Documentation of shop drawings including date submitted to and date approved by the contracting officer, results of examinations, any need for changes or modifications, manufacturer's recommendations and certifications, if any, and signature of the authorized examiner.

- b. Documentation of material delivered including quantity, storage location, and results of quality control examinations and tests.
- c. Type, number, date, time, and name of individual performing quality control activities.
- d. The material or item inspected and tested, the location and extent of such material or item, and a description of conditions observed and test results obtained during the quality control activity.
- e. The determination that the material or item met the contract provisions and documentation that the engineer was notified.
- f. For deficient work, the nature of the defects, specifications not met, corrective action taken, and results of quality control activities on the corrected material or item.

7. Reporting results

The results of contractor quality control inspections and tests shall be communicated to the engineer immediately upon completion of the inspection or test. Unless otherwise specified in section 10, the original plus one copy of all records, inspections, tests performed, and material testing reports shall be submitted to the engineer within one working day of completion. The original plus one copy of documentation of material delivered shall be submitted to the engineer before the material is used.

8. Access

The contracting officer and the engineer shall be given free access to all testing equipment, facilities, sites, and related records for the duration of the contract.

9. Payment

Method 1—For items of work for which lump sum prices are established in the contract, payment is made as the work proceeds, after presentation by the contractor of invoices showing related costs and evidence of charges by suppliers, subcontractors, and others for furnishing supplies and work performed. If the total of such payments is less than the lump sum contract price for this item, the remaining balance is included in the final contract payment. Payment of the lump sum contract price constitutes full compensation for completion of the work.

Payment is not made under this item for the purchase cost of material and equipment having a residual value.

Method 2—For items of work for which lump sum prices are established in the contract, payment is prorated and paid in equal amounts on each monthly estimate. The number of months used for prorating shall be the number estimated to complete the work. The final month's prorate amount is made with the final payment. Payment as described above constitutes full compensation for completion of the work.

Payment is not made under this item for the purchase cost of material and equipment having a residual value.

All methods—Compensation for any item of work described in the contract, but not listed in the bid schedule, is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in section 10.

10. Items of work and construction details

Items of work to be performed in conformance with this specification and the construction details therefore are:

1. In Section 3 - Quality control system, Method 1 applies. Verbal notification of the quality control personnel submitted at the beginning of activities is required.
2. In Section 4 – Quality control personnel, Method 1 applies. The project supervisor may act as the contractor's quality control manager.
3. Section 9 Payment does not apply. No separate payment will be made for Contractor Quality Control. Compensation for this item will be included in the payment for Bid Item 1 - Mobilization and Demobilization.

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Construction Specification 95 — Geotextile

Title 210 – Engineering

Part 642 – Specifications for Construction Contracts

Subpart B – National Construction Specifications

June 2024

A. Scope

1. This work consists of furnishing all material, equipment, and labor necessary for the installation of geotextiles.

B. Quality

1. Geotextiles must conform to the requirements of Material Specification 592 and this specification.

C. Storage

1. Before use, store the geotextile in a clean, dry location out of direct sunlight, not subject to extremes of either hot or cold temperatures, and with the manufacturer's protective cover undisturbed. Receiving, storage, and handling at the jobsite must be in accordance with the requirements listed in ASTM D4873.

D. Surface preparation

1. The surface on which the geotextile is to be placed must be graded to the neat lines and grades as shown on the drawings. The surface must be reasonably smooth and free of loose rock and clods, holes, depressions, projections, muddy conditions, and standing or flowing water (unless otherwise specified in section G of this specification).

E. Placement

1. Before the geotextile is placed, the soil surface will be reviewed for quality assurance of the design and construction. Place the geotextile on the approved prepared surface at the locations and in accordance with the details shown on the drawings and specified in section G of this specification. It must be unrolled along the placement area and loosely laid, without stretching, in such a manner that it conforms to the surface irregularities when material or gabions are placed on or against it. The geotextile may be folded and overlapped to permit proper placement in designated areas.
2. Method 1.
 - a. Join the geotextile by machine sewing using thread material meeting the chemical requirements for the geotextile fibers or yarn. Thread must be polypropylene, polyester, or Kevlar™ aramid thread unless a specific thread type is specified. The thread must consist of two parallel stitched rows at a

spacing of about 1 inch and must not cross (except for any required re-stitching). The stitching must be a lock-type stitch. Each row of stitching must be located a minimum of 2 inches from the geotextile edge. Unless otherwise specified, the seam tensile strength as measured according to ASTM D4884 must be a minimum of 90 percent of the geotextile tensile strength in the weakest principal direction as measured according to ASTM D4632.

- b. The geotextile must be temporarily secured during placement of overlying material to prevent slippage, folding, wrinkling, or other displacement of the geotextile. Unless otherwise specified, methods of securing must not cause punctures, tears, or other openings to be formed in the geotextile.
3. Method 2.
 - a. Join the geotextile by overlapping a minimum of 18 inches (unless otherwise specified) and secured against the underlying foundation material. Securing pins, approved and provided by the geotextile manufacturer, must be placed along the edge of the panel or roll material to adequately hold it in place during installation. Pins must be steel or fiberglass formed as a U, L, or T shape or must contain ears to prevent total penetration through the geotextile. Provide steel washers on all but the U-shaped pins. The upstream or upslope geotextile must overlap the abutting downslope geotextile. At vertical laps, securing pins must be inserted through the bottom layers along a line through approximately the midpoint of the overlap. At horizontal laps and across slope laps, securing must be inserted through the bottom layer only. Securing pins must be placed along a line about 2 inches in from the edge of the placed geotextile at intervals not to exceed 12 feet unless otherwise specified. Install additional pins as necessary and where appropriate to prevent any undue slippage or movement of the geotextile. The use of securing pins will be held to the minimum necessary. Pins are to remain in place unless otherwise specified.
 - b. Should the geotextile be torn or punctured or the overlaps or sewn joint disturbed as evidenced by visible geotextile damage, subgrade pumping, intrusion, or grade distortion, the backfill around the damaged or displaced area must be removed and restored to the original approved condition. The repair must consist of a patch of the same type of geotextile being used and overlaying the existing geotextile. When the geotextile seams are required to be sewn, extend the overlay patch a minimum of 1 foot beyond the edge of any damaged area and join by sewing as required for the original geotextile except that the sewing must be a minimum of 6 inches from the edge of the damaged geotextile. Geotextile panels joined by overlap must have the patch extend a minimum of 2 feet from the edge of any damaged area.
 4. Place geotextile in accordance with the following applicable specification according to the use indicated in section G:
 - a. Slope protection. The geotextile must not be placed until it can be anchored and protected with the specified covering within 48 hours or protected from exposure to ultraviolet light. In no case may material be dropped on uncovered geotextile from a height of more than 3 feet.

- b. Subsurface drains. The geotextile must not be placed until drainfill or other material can be used to provide cover within the same working day. Drainfill material must be placed in a manner that prevents damage to the geotextile. In no case may material be dropped on uncovered geotextile from a height of more than 5 feet.
- c. Road stabilization. The geotextile must be unrolled in a direction parallel to the roadway centerline in a loose manner permitting conformation to the surface irregularities when the roadway fill material is placed on its surface. In no case may material be dropped on uncovered geotextile from a height of more than 5 feet. Unless otherwise specified, the minimum overlap of geotextile panels joined without sewing must be 24 inches. The geotextile may be temporarily secured with pins recommended or provided by the manufacturer, but they must be removed before the permanent covering material is placed.

F. Measurement and payment

- 1. Method 1. For items of work for which specific unit prices are established in the contract, the quantity of geotextile for each type placed within the specified limits is determined to the nearest specified unit by measurements of the covered surfaces only, disregarding that required for anchorage, seams, and overlaps. Payment is made at the contract unit price. Such payment constitutes full compensation for the completion of the work.
- 2. Method 2. For items of work for which specific unit prices are established in the contract, the quantity of geotextile for each type placed within the specified limits is determined to the nearest specified unit by computing the area of the actual roll size or partial roll size installed. The computed area will include the amount required for overlap, seams, and anchorage as specified. Payment is made at the contract unit price. Such payment constitutes full compensation for the completion of the work.
- 3. Method 3. For items of work for which specific lump sum prices are established in the contract, the quantity of geotextile is not measured for payment. Payment for geotextiles is made at the contract lump sum price and constitutes full compensation for the completion of the work.
- 4. All Methods. The following provisions apply to all methods of measurement and payment. Compensation for any item of work described in the contract but not listed in the bid schedule is included in the payment for the item of work to which it is made subsidiary. Such items and the items to which they are made subsidiary are identified in section G of this specification.

G. Items of work and construction details

Subsidiary Item – Nonwoven Geotextile Fabric

1. This item shall consist of furnishing and placing geotextile fabric under the drainfill placed behind the sheet pile wall as per the drawings.
2. Geotextile fabric shall meet the material specifications of Class III non-woven geotextile as outlined in Material Specification 592 – Geotextile.
3. Seams or joints in the geotextile fabric shall be installed per Method 2 and have 18” of overlap.
4. Section F Measurement and Payment does not apply. No separate payment will be made for Subsidiary Item – Nonwoven Geotextile Fabric. Compensation for this item will be included in the payments for Bid Item 6 – Drainfill.

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Material Specification 511—Steel Piles

1. Scope

The specification covers the type and quality of steel piles.

2. Bearing piles

Steel bearing piles shall be structural steel H-piles conforming to the requirements of ASTM A36.

The required length of pile may be fabricated by butt-welding shorter lengths of pile stock. Unless otherwise specified, the cross-section of each pile shall be constant throughout its length. The axis of the pile shall be straight, and the number of welded joints in the length of the pile shall be as few as practicable. Pieces below the top piece shall have a minimum length of 10 feet.

3. Sheet piles

Steel sheet piles shall conform to the requirements of ASTM A328, A572, or A690. Fabrication of piles from shorter lengths of pile stock is not permitted.

Material Specification 521—Aggregates for Drainfill and Filters

1. Scope

This specification covers the quality of mineral aggregates for the construction of drainfill and filters.

2. Quality

Drainfill and filter aggregates shall be sand, gravel, or crushed stone or mixtures thereof. Aggregates shall be composed of clean, hard, durable, mineral particles free from organic matter, clay balls, soft particles, or other substances that would interfere with the free-draining properties of the aggregates.

Coarse aggregate may be crushed limestone or other material that has limestone particles included. Aggregates from crushed limestone shall be thoroughly washed and screened to remove limestone dust, limestone fines, and fine soil particles. Limestone shall not be used for fine aggregates except in combination with other material, such that not more than 5 percent of the portion finer than the No. 4 sieve shall be limestone.

Aggregates shall be tested for soundness according to ASTM Method C 88 and shall have a weighted average loss in 5 cycles of not more than 12 percent when sodium sulfate is used or 18 percent when magnesium sulfate is used.

3. Grading

Drainfill and filter aggregates shall conform to the specified grading limits after being placed or after being compacted when compaction is specified. Grading shall be determined by ASTM Method C 136. The percentage of material finer than the No. 200 sieve shall be determined by the method in ASTM Designation C 117.

4. Storing and handling

Drainfill and filter aggregates shall be stored and handled by methods that prevent segregation of particle sizes or contamination by mixing with other material.

Material Specification 523—Rock for Riprap

1. Scope

This specification covers the quality of rock to be used in the construction of rock riprap.

2. Quality

Individual rock fragments must be dense, sound, and free from cracks, seams, and other defects conducive to accelerated weathering. Except as otherwise specified, the rock fragments must be angular to subrounded. The least dimension of an individual rock fragment must not be less than one-third the greatest dimension of the fragment. ASTM D4992 provides guidance on selecting rock.

Except as otherwise provided, the rock must be tested and must have the following properties:

Rock type 1

- **Bulk specific gravity (saturated surface-dry basis)**—Not less than 2.5 when tested in accordance with ASTM D6473 on samples prepared as described for soundness testing.
- **Absorption**—Not more than 2 percent when tested in accordance with ASTM D6473 on samples prepared as described for soundness testing.
- **Soundness**—The weight loss in five cycles must not be more than 10 percent when sodium sulfate is used or more than 15 percent when magnesium sulfate is used.

Rock type 2

- **Bulk specific gravity (saturated surface-dry basis)**—Not less than 2.5 when tested in accordance with ASTM D6473 on samples prepared as described for soundness testing.
- **Absorption**—Not more than 2 percent when tested in accordance with ASTM D6473 on samples prepared as described for soundness testing.
- **Soundness**—The weight loss in five cycles must be not more than 20 percent when sodium sulfate is used or more than 25 percent when magnesium sulfate is used.

Rock type 3

- **Bulk specific gravity (saturated surface-dry basis)**—Not less than 2.3 when tested in accordance with ASTM D6473 on samples prepared as described for soundness testing.
- **Absorption**—Not more than 4 percent when tested in accordance with ASTM D6473 on samples prepared as described for soundness testing.
- **Soundness**—The weight loss in five cycles must be not more than 20 percent when sodium sulfate is used or more than 25 percent when magnesium sulfate is used.

3. Methods of soundness testing

Rock cube soundness—The sodium or magnesium sulfate soundness test for all rock types (1, 2, or 3) must be performed on a test sample of $5,000 \pm 300$ grams of rock fragments, reasonably uniform in size and cubical in shape, and weighing, after sampling, about 100 grams each. They must be obtained from rock samples that are representative of the total rock mass, as noted in ASTM D4992, and that have been sawed into slabs as described in ASTM D5121. The samples must further be reduced in size by sawing the

slabs into cubical blocks. The thickness of the slabs and the size of the sawed fragments must be determined by the size of the available test apparatus and as necessary to provide, after sawing, the approximate 100-gram samples. The cubes must undergo five cycles of soundness testing in accordance with ASTM C88.

Internal defects may cause some of the cubes to break during the sawing process or during the initial soaking period. Do not test any of the cubes that break during this preparatory process. Such breakage, including an approximation of the percentage of cubes that break, must be noted in the test report.

After the sample has been dried following completion of the final test cycle and washed to remove the sodium sulfate or magnesium sulfate, the loss of weight must be determined by subtracting from the original weight of the sample the final weight of all fragments that have not broken into three or more fragments.

The test report must show the percentage loss of the weight and the results of the qualitative examination.

Rock slab soundness—When specified, the rock must also be tested in accordance with ASTM D5240. Deterioration of more than 25 percent of the number of blocks is cause for rejection of rock from this source. Rock must also meet the requirements for average percent weight loss stated below.

- For projects located north of the Number 20 Freeze-Thaw Severity Index Isoline (fig. 523–1), unless otherwise specified, the average percent weight loss for Rock Type 1 must not exceed 20 percent when sodium sulfate is used or 25 percent when magnesium sulfate is used. For Rock Types 2 and 3, the average percent weight loss must not exceed 25 percent for sodium sulfate soundness or 30 percent for magnesium sulfate soundness.
- For projects located south of the Number 20 Freeze-Thaw Severity Index Isoline, unless otherwise specified, the average percent weight loss for Rock Type 1 must not exceed 30 percent when sodium sulfate is used or 38 percent when magnesium sulfate is used. For Rock Types 2 and 3, the average percent weight loss must not exceed 38 percent for sodium sulfate soundness or 45 percent for magnesium sulfate soundness.

Figure 523–1

Number 20 Freeze-Thaw Severity Index Isoline (map approximates the map in ASTM D5312)



4. Field durability inspection

Rock that fails to meet the material requirements stated above (if specified), may be accepted only if similar rock from the same source has been demonstrated to be sound after 5 years or more of service under conditions of weather, wetting and drying, and erosive forces similar to those anticipated for the rock to be installed under this specification.

A rock source may be rejected if the rock from that source deteriorates in less than 5 years under similar use and exposure conditions expected for the rock to be installed under this specification, even though it meets the testing requirements stated above.

Deterioration is defined as the loss of more than one-quarter of the original rock volume, or severe cracking that would cause a block to split. Measurements of deterioration are taken from linear or surface area particle counts to determine the percentage of deteriorated blocks. Deterioration of more than 25 percent of the pieces is cause for rejection of rock from the source.

5. Grading

The rock must conform to the specified grading limits after it has been placed within the matrix of the rock riprap. Grading tests must be performed, as necessary, according to ASTM D5519, Method A, B, or C, as applicable.

Part 642 – Specifications

Chapter 3 – National Standard Material Specifications

Material Specification 547—Plastic Pipe

A. Scope

This specification covers the quality of Poly Vinyl Chloride (PVC), Polyethylene (PE), High Density Polyethylene (HDPE), and Acrylonitrile-Butadiene-Styrene (ABS) plastic pipe, fittings, and joint materials.

B. Material

- (1) Pipe—The pipe must be as uniform as commercially practicable in color, opaqueness, density, and other specified physical properties. It must be free from visible cracks, holes, foreign inclusions, or other defects. The dimensions of the pipe must be measured as prescribed in ASTM D2122. Unless otherwise specified, the pipe must conform to the requirements listed in this specification and the applicable reference specifications in Figure 547–2, the requirements specified in Construction Specification 45, Plastic Pipe, and the requirements shown on the drawings.
- (2) Fittings and joints—Fittings and joints must be of a schedule, SDR or DR, pressure class, external load carrying capacity, or pipe stiffness that equals or exceeds that of the plastic pipe. The dimensions of fittings and joints must be compatible with the pipe and measured in accordance with ASTM D2122. Joint and fitting material must be compatible with the pipe material. The joints and fittings must be as uniform as commercially practicable in color, opaqueness, density, and other specified physical properties. It must be free from visible cracks, holes, foreign inclusions, or other defects. Fittings and joints must conform to the requirements listed in this specification, the requirements of the applicable specification referenced in the ASTM or AWWA specification for the pipe, the requirements specified in Construction Specification 45, and the requirements shown on the drawings.
- (3) Solvents—Solvents for solvent-welded pipe joints must be compatible with the plastic pipe used and must conform to the requirements of the applicable specification referenced in the ASTM or AWWA specification for the pipe, fitting, or joint.
- (4) Gaskets—Rubber gaskets for pipe joints must conform to the requirements of ASTM F477, Elastomeric Seals (Gaskets) for Jointing Plastic Pipe.

C. Perforations

When perforated pipe is specified, perforations must conform to the following requirements unless otherwise specified in Construction Specification 45 or shown on the drawings:

- (1) Perforations must be either circular or slots.
- (2) Circular perforations must be $1/4 \pm 1/16$ -inch diameter holes arranged in rows parallel to the axis of the pipe.
- (3) Perforations must be evenly spaced along each row such that the center-to-center distance between perforations is not less than eight times the perforation diameter. Perforations may appear at the ends of short and random lengths. The minimum perforation opening per foot of pipe must be as shown in Figure 547–1.

Figure 547-1 Perforations

Nominal pipe size	Minimum number of rows		Minimum opening/foot (in ²)
	circular	slot	
4	2	2	0.22
6	4	2	0.44
8	4	2	0.44
10	4	2	0.44
12	6	2	0.66

- (4) Rows must be arranged in two equal groups at equal distance from the bottom on each side of the vertical centerline of the pipe. The lower-most rows of perforations must be separated by an arc of not less than 60 degrees or more than 125 degrees. The uppermost rows of perforations must be separated by an arc not to exceed 166 degrees. The spacing of rows between these limits must be uniform. The minimum number of rows must be as shown in Figure 547–1.
- (5) Slot perforations must be symmetrically located in two rows, one on each side of the pipe centerline. Slot perforations must be located within the lower quadrants of the pipe with slots no wider than 1/8 inch and spaced not to exceed 11 times the perforation width. Minimum perforation opening per lineal foot of pipe must be as shown in Figure 547–1.
- (6) On both the inside and outside of the pipe, perforations must be free of cuttings or frayed edges and of any material that would reduce the effective opening.

Figure 547-2 Pipe Specifications

Pipe	Specification
Poly vinyl chloride (PVC) pipe	
Plastic pipe - Schedules 40, 80, 120	ASTM D1785 ASTM D2466
Pressure rated pipe - SDR Series	AWWA C900 ASTM D2241
Plastic drain, waste, and vent pipe and fittings	ASTM D2665
Joints for IPS PVC pipe using solvent weld cement	ASTM D2672
Composite sewer pipe	ASTM D2680
Type PSM PVC sewer pipe and fittings	ASTM D3034
Large-diameter gravity sewer pipe and fittings	ASTM F679
Smooth-Wall Underdrain Systems for Highway, Airport, and Similar Drainage	ASTM F758
Profile gravity sewer pipe and fittings based on controlled inside diameter	ASTM F794
Corrugated sewer pipe with a smooth	ASTM F949

Title 210 – National Engineering Handbook

Pipe	Specification
interior and fittings	
Pressure pipe, 4-inch through 60-inch for water distribution	AWWA C900
Polyethylene (PE) plastic pipe	
12- to 60-inch annular corrugated profile-wall polyethylene (PE) pipe and fittings	ASTM F2306
SIDR-PR based on controlled inside diameter	ASTM D2239
SDR-PR based on controlled outside diameter	ASTM D3035
High density polyethylene (HDPE) plastic pipe	
Plastic pipe and fittings	ASTM D3350
SDR-PR based on controlled outside diameter	ASTM F714
Heat joining polyolefin pipe and fittings	ASTM D2657
Acrylonitrile-butadiene-styrene (ABS) pipe	
Composite sewer pipe	ASTM D2680

Part 642 – Specifications

Chapter 3 – National Standard Material Specifications

Material Specification 592—Geotextile

A. Scope

This specification covers the quality of geotextile, including geotextile for temporary silt fence.

B. General Requirements

- (1) Fibers (threads and yarns) used in the manufacture of geotextile must consist of synthetic polymers composed of a minimum of 85 percent by weight polypropylene, polyester, polyamide, polyethylene, polyolefin, or polyvinylidene-chloride. The fiber must be formed into a stable network of filaments retaining dimensional stability relative to each other. The geotextile must be free of defects, such as holes, tears, and abrasions. The geotextile must be free of any chemical treatment or coating that significantly reduces its porosity. Fibers must contain stabilizers, inhibitors, or both to enhance resistance to ultraviolet light. Geotextile other than for temporary silt fence must conform to the requirements in Figure 592-1 or 592-2, as applicable. Geotextile for temporary silt fence must conform to ASTM D6461.
- (2) Thread used for factory or field sewing must be of a color contrasting to the color of the fabric and made of high-strength polypropylene, polyester, or polyamide material. It must be as resistant to ultraviolet light as the geotextile being sewn.

C. Classification

- (1) There are two geotextile classifications, woven and nonwoven. Geotextile for temporary silt fence may be either woven or nonwoven. Silt film woven geotextile may not be used except for temporary silt fence.
- (2) Woven geotextiles are made from fabric that is formed by the uniform and regular interweaving of the threads or yarns in two directions. Woven fabrics must be manufactured from monofilament yarn formed into a uniform pattern with distinct and measurable openings, retaining their position relative to each other. The fabric must have a selvage edge or otherwise be finished to prevent unraveling.
- (3) Nonwoven geotextiles are made from fabric that is formed by a random placement of threads in a mat and bonded by needle punching, heat bonding, or resin bonding. Nonwoven geotextiles must have distinct but variable small openings, retaining their position relative to each other when bonded. The use of heat- or resin-bonded nonwovens is restricted as specified in note 2 of Figure 592-2.

D. Sampling and Testing

The geotextile must conform to Figure 592-1 or 592-2 or ASTM D6461, as applicable, for the product type shown on the label. Documentation described in either (i) or (ii) below is required to verify the product meets the specified requirements:

- (i) Product properties as listed in the latest edition of the "Specifiers Guide," Geosynthetics (Industrial Fabrics Association International, 1801 County Road B, West Roseville, MN 55113-4061 or at <http://www.geosindex.com>), and that represent average roll values, are acceptable.

- (ii) Test data from the geotextile production run for each of the specified tests listed in Figure 592-1 or 592-2 or ASTM D6461, as applicable.

E. Shipping and Storage

Each roll of geotextile must be labeled or tagged to clearly identify the brand, class, and individual production run in accordance with ASTM D4873. The geotextile must be shipped and transported in rolls wrapped with a cover for protection from moisture, dust, dirt, debris, and ultraviolet light. The cover must be maintained undisturbed to the maximum extent possible before placement.

Figure 592-1 Requirements for Woven Geotextiles ^{1/}

Property	Test method	Class I	Class II	Class III	Class IV
Grab tensile strength (lb)	ASTM D4632	247 minimum	180 minimum	180 minimum	315
Elongation at failure (%)	ASTM D4632	<50	<50	<50	<50
Trapezoidal tear strength (lb)	ASTM D4533	90 minimum	67 minimum	67 minimum	112 minimum
Puncture strength (lb)	ASTM D6241	495 minimum	371 minimum	371 minimum	618 minimum
Ultraviolet stability (% retained strength)	ASTM D4355	50 minimum	50 minimum	50 minimum	70 minimum
Permittivity (sec ⁻¹)	ASTM D4491	as specified			
Apparent opening size (AOS) ^{2/}	ASTM D4751	as specified			
Percent open area (POA) (%)	USACE ^{3/} CWO-02215-86	as specified			

^{1/} All values are minimum average roll values (MARV) in the weakest principal direction, unless otherwise noted.

^{2/} Maximum average roll value.

^{3/} Note: CWO is a USACE reference.

Figure 592-2 Requirements for Non-Woven Geotextiles ^{1/}

Property	Test method	Class I ^{2/}	Class II ^{2/}	Class III ^{2/}	Class IV ^{2/}
Grab tensile strength (lb)	ASTM D4632 grab test	202 minimum	157 minimum	112 minimum	202 minimum
Elongation at failure (%)	ASTM D4632	50 minimum	50 minimum	50 minimum	50 minimum
Trapezoidal tear strength (lb)	ASTM D4533	79 minimum	56 minimum	40 minimum	79 minimum
Puncture strength (lb)	ASTM D6241	433 minimum	309 minimum	223 minimum	433 minimum
Ultraviolet light (% retained strength)	ASTM D4355	50 minimum	50 minimum	50 minimum	50 minimum
Permittivity sec ⁻¹	ASTM D4491	0.70 minimum or as specified			
Apparent opening size (AOS) (mm) ^{3/}	ASTM D4751	0.22 maximum or as specified			

^{1/} All values are minimum average roll values (MARV) in the weakest principal direction, unless otherwise noted.

^{2/} Needle punched geotextiles may be used for all classes. Heat-bonded or resin-bonded geotextiles may be used for class IV only.

^{3/} Maximum average roll value.